The following is an excerpt from the doctoral dissertation:

Davis, E. M. (2026). *Investigating cell shape, motion, and force generation over time scales relevant for mesenchymal migration*. University of North Carolina at Chapel Hill.

CHAPTER III: MODELS OF CELL MIGRATION

3.1 Introduction

The ability for cells to migrate, both individually and collectively, is crucial in a wide array of physiological processes (Fortunato and Sunyer, 2022; Petrie et al., 2009; SenGupta et al., 2021b; Shellard and Mayor, 2020; Sunyer and Trepat, 2020; Wu et al., 2014). For instance, cell migration is important in immune cells surveillance and recruitment to sites of infection (Vesperini et al., 2021). In embryogenesis, neural crest cells must be able to migrate long distances to contribute to the development of a variety of tissues (Shellard and Mayor, 2019). Cell migration is not only prevalent in processes vital to development and maintaining homeostasis, but also in malignant processes such cancer metastasis (Brückner and Broedersz, 2024b; Shellard and Mayor, 2020). Typically, cells travel to necessary locations due to external cues, both chemical and mechanical. However, in the absence of strong external signals, motile cells undergo random cell migration which allows them to explore their environment (Petrie et al., 2009; SenGupta et al., 2021b; Shellard and Mayor, 2020).

While mesenchymal cell motility is relevant in a broad array of biological processes amongst many different cell types, the machinery itself is highly conserved (Brückner and Broedersz, 2024b). For movement to occur, the cell must be polarized with a defined front and back. Actin polymerization occurs at the leading edge of the cell forming protrusions. Within

these protrusions, adhesions begin to form between the cell and substrate. These contact sites are also where bundles of actin and non-muscle myosin II motors attach and generate contractile forces that cause the adhesion to mature, thus forming a stronger connection with the extracellular matrix (ECM) and resulting in traction (Shellard and Mayor, 2020). Conversely, at the cell rear adhesions disassemble and the cell body and nucleus retract. Taken together, the spatial organization of forces generated by actin polymerization and myosin contraction is what ultimately what results in net cellular displacement (Shellard and Mayor, 2020). Due to the crucial roles cell migration plays in a variety of processes, it is often helpful to quantify the dynamics of migrating cells to incorporate into a model to make predictions about observed behaviors.

3.2 Persistent Random Walk

The Langevin Equation

One of the simplest ways to quantitatively represent cell migration is by modelling the position of a cell as a function of time. One model that is used for this is the persistent random motion model, given by the Langevin equation, a stochastic differential equation (Brückner and Broedersz, 2024b; Uwamichi et al., 2023):

$$\frac{d\vec{v}(t)}{dt} = -\beta\vec{v}(t) + \sigma\vec{\eta}(t)$$

Where $\vec{v} = d\vec{x}/dt$ is the cell velocity, $\vec{\eta}(t)$ is 2D Gaussian white noise, and σ is the strength of the noise. The persistence time, β^{-1}, is the decay time over cell velocity correlations. While this representation is helpful to get a general idea of cell motility behavior, it does not give

an accurate representation of observed behaviors. For example, we expect the velocity autocorrelation to follow a single exponential decay, $\langle v(\tau)v(0)\rangle \sim e^{-\beta\tau}$. However, when we attempt to fit the velocity autocorrelation function for a population of cells undergoing random migration, we find that a single exponential fit fails to accurately capture the data (Fig. 3.1). This is particularly evident in the *Arpc*2 KO cells, which display a more pronounced longer time scale decay.

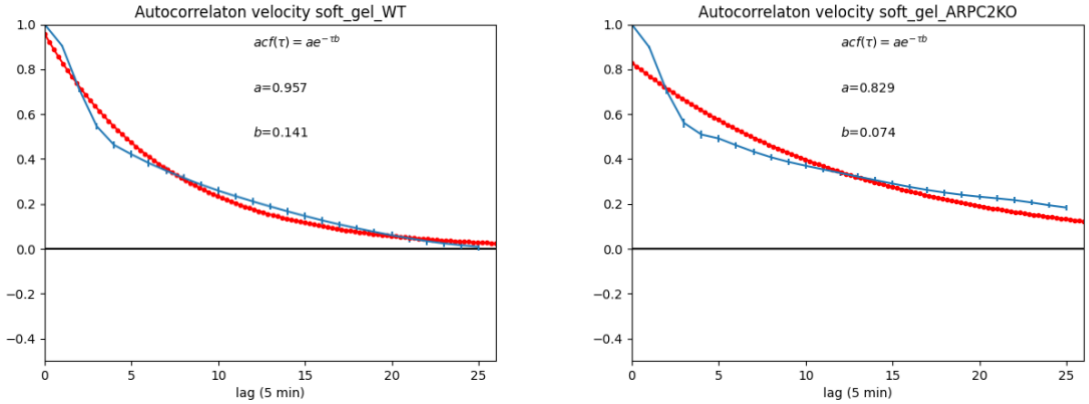


Figure 3.1. The fit of a single exponential decay to velocity ACF data.

The velocity autocorrelation function from the Langevin equation has the form of a single exponential decay. Fitting an equation of this form to velocity autocorrelation data of WT and *Arpc*2 KO cells does not accurately capture the observations.

The Generalized Langevin Equation

According to the Langevin equation, it is expected that the distribution of velocity is Gaussian and the velocity autocorrelation function has a single exponential decay. However, as discussed previously, this is not what is frequently observed experimentally. Typically, the velocity distributions exhibit exponential tails and the velocity autocorrelation function best matches the sum of two exponential functions, thus suggesting two decay times are necessary for

describing the position of cells over time (Uwamichi et al., 2023; Wu et al., 2014). The simplest model consistent with these observed statistics is given by the generalized Langevin equation, which includes a memory kernel dependent on past velocities:

$$\frac{d\vec{v}(t)}{dt} = -\beta\vec{v}(t) + \alpha^2 \int_{-\infty}^{t} e^{-\gamma(t-t')}\vec{v}(t')dt' + \sigma\vec{\eta}(t)$$

Where the second time scale is given by memory time scale γ^{-1}. According to this equation, we expect the velocity autocorrelation to decay as the sum of two different time scales (Uwamichi et al., 2023). In fact, when we fit experimental data of cells undergoing random migration, we find that the double exponential fit closely follows our data (Fig. 3.2).

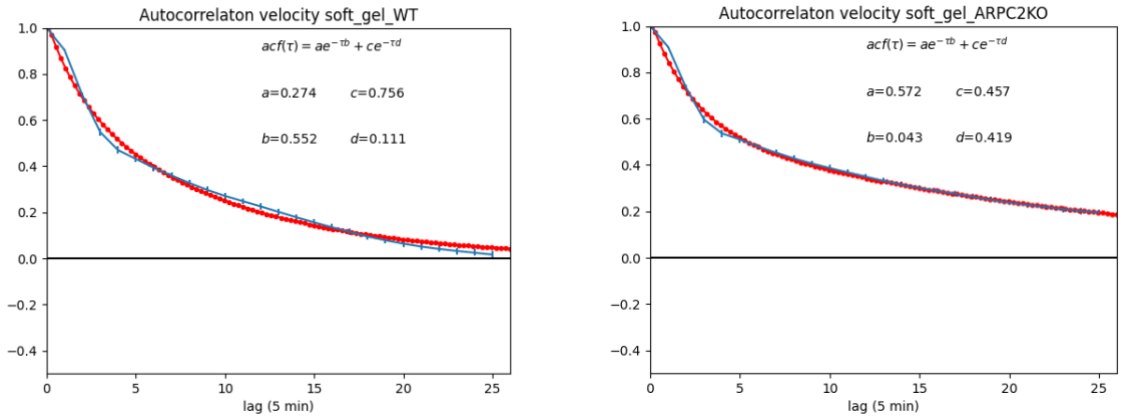


Figure 3.2. The fit of two exponential decays to velocity ACF data.

The velocity autocorrelation function from the generalized Langevin equation has the form of the sum of two exponential decays. Fitting an equation of this form to velocity autocorrelation data of WT and *Arpc*2 KO cells provides a much better fit than that of the Langevin equation.

3.3 Vertex Based Model

While the generalized Langevin equation is helpful for capturing behavior of overall cell positions over time, it treats the cell as a single point and excludes spatial information of activity occurring over the scale of the entire cell. We turn to a vertex based model (VBM) to represent a single cell to formulate a model that aims to propose mechanisms for how shape, traction, and motion are related and the time scales over which they occur.

Model Setup and Relevant Forces

A single cell is represented as a polygon with N discrete points (vertices) around the perimeter. The vertices have a linear elastic interaction with each other (Fig. 3.3), where the elastic tension along the edge between vertices i and $i+1$ are given by the following equation:

$$\vec{F}_{edge_i} = \lambda \frac{\Delta l_i}{l_0}$$

Here, λ is the membrane elasticity, Δl_i is the change in length from the undeformed length, and l_o is the undeformed length.

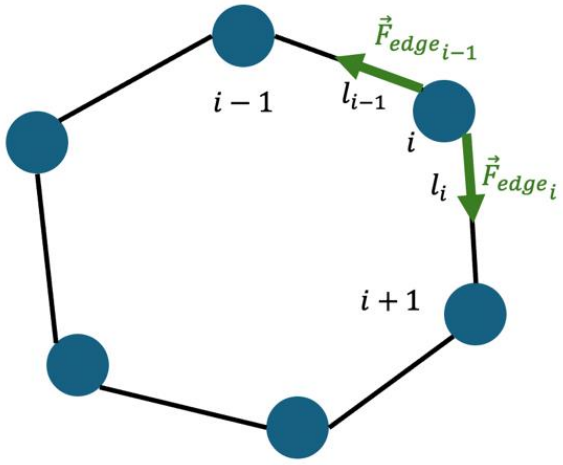


Figure 3.3. Linear elastic force between vertices.

There is a linear elastic force, $\vec{F}_{edge_i}$, between each vertex, i, and its neighbors $i+1$ and $i-1$.

The cell is assumed to be at rest when the vertices are equally spaced from each other. The displacement of a vertex leads to a change in edge length from the undeformed length, thus resulting in a force along each edge connecting to the displaced vertex.

Additionally, there is a pressure that is exerted on each vertex that acts in the normal direction and works to maintain the area of the cell. This cytoplasmic pressure at vertex i is given by:

$$p_i = \left(k_c\left[1 - \frac{A(t)}{A_0}\right]\right)/N$$

Where k_c is the cytoplasmic stiffness, $A(t)$ is the cell area at time t, A_0 is the resting cell area, and N is the number of vertices. This pressure acts in the unit outward normal vector at vertex i, denoted as $\vec{n}_i$ (Fig. 3.4).

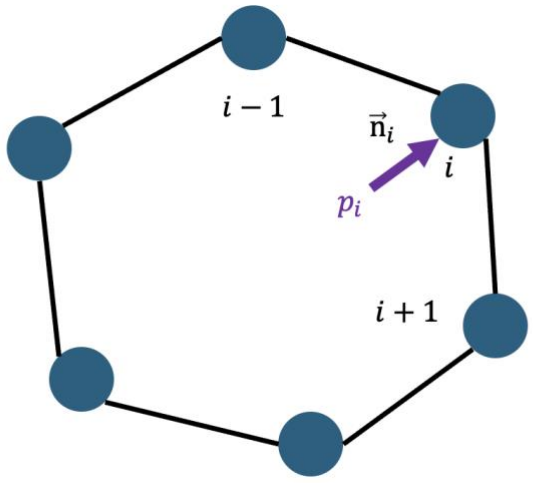


Figure 3.4. Cytoplasmic pressure.

There is an outward normal force acting on each vertex representing the cytoplasmic pressure.

Thus, a decrease in cell area from rest cell area results in a normal outward pressure force at each vertex. Conversely, an increase in cell area from rest cell area results in a normal inward force at each vertex.

In addition to the spring forces between vertices and the normal pressure force, there is third force that acts normal outward at vertices: a stochastic protrusive force that turns on with rate k_{on} and turns off with rate k_{off}. We will cover this stochastic protrusive force in more depth in the next section.

To summarize: the shape of a single cell is represented by a polygon with discrete locations on the edge called vertices. There are three different types of forces that can act on each vertex: 1) A simple linear spring force that acts along the edge connecting a vertex to its neighbors on either side. 2) A normal pressure force that acts to maintain cell area. 3) A stochastic protrusive force that acts normal outward.

On and Off Rates for the Stochastic Protrusive Force, F_{pr}

As stated previously, there is a stochastic protrusive force, F_{pr}, that acts normally outward in the $\vec{n}_i$ direction that turns on and off with rates k_{on} and k_{off}. The stochastic protrusive force acts at vertex i at time t with the following equation:

$$F_{pr}(t)_i = \begin{cases} F_{on} = constant \\ F_{off} = 0 \end{cases}$$

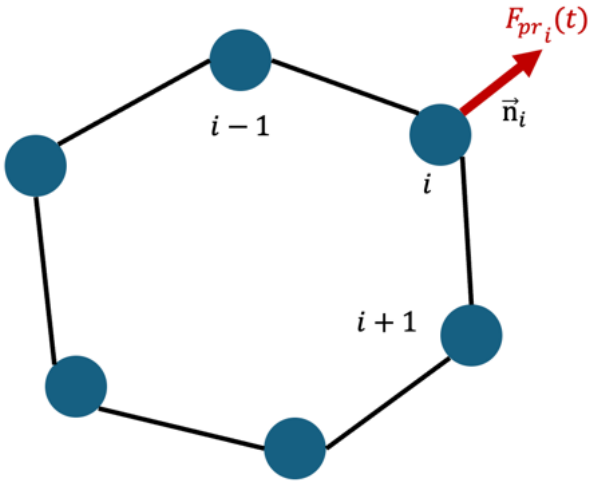


Figure 3.5. Stochastic protrusive force.

There is a protrusive force, F_{pr}, that can stochastically turn on and off at each vertex (Fig. 3.5).

On Rate, k_{on}

The rate by which the stochastic protrusive force turns on for each vertex is affected by the sum of the linear elastic spring forces along the edges to the neighboring vertices projected onto the normal outward axis (Fig. 3.6), given by the equation:

$$T_i = \left(\vec{F}_{edge_{i-1}} + \vec{F}_{edge_i}\right) \cdot \vec{n}_i$$

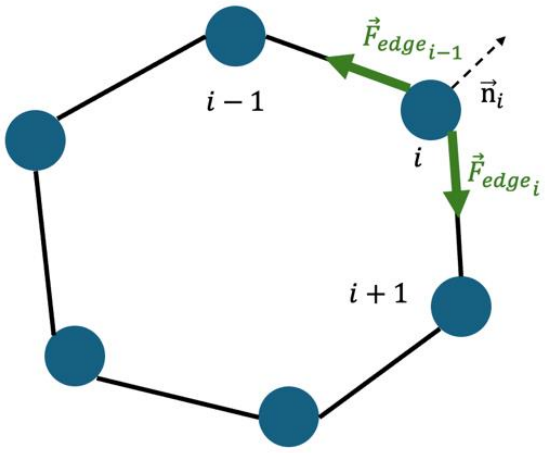


Figure 3.6. Tension due to elastic forces.

There is a tension acting in the outward normal direction at each vertex due to the linear elastic force, $\vec{F}_{edge_i}$, between the vertices.

We will refer to this quantity, T_i, as the tension at vertex i. A negative value of T_i means that the sum of $(\vec{F}_{edge_{i-1}} + \vec{F}_{edge_i})$ faces in towards the center of cell, thus indicating the vertex is displaced further outwards and therefore likely protruding. Conversely, a positive value of T_i means that the sum of $(\vec{F}_{edge_{i-1}} + \vec{F}_{edge_i})$ is pointing outwards from the cell center and its neighboring vertices are likely to be protruding while this vertex is not. Using this quantity, we can calculate the on rate, k_{on} for F_{pr} at vertex i using:

$$k_{on_i}(T_i) = k_{on}^{*} e^{T_i/T_{max}}$$

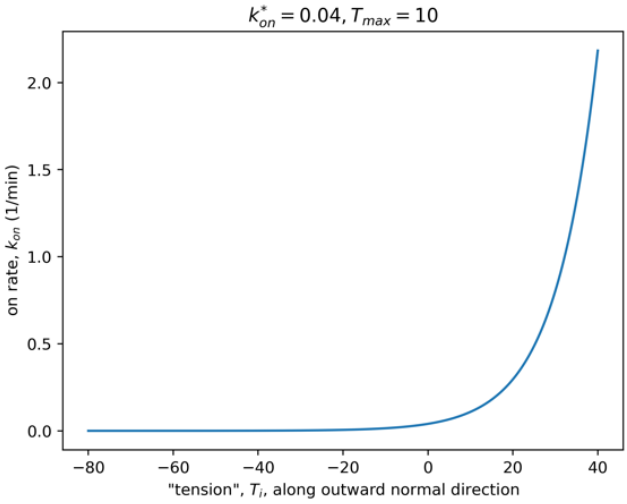


Figure 3.7. On rate for stochastic protrusive force.

Example of how the on rate for the stochastic protrusive force, F_{pr}, varies with tension, T, using values of $k^*_{on} = 0.04\ min^{-1}$ and $T_{max} = 10\ nN$.

As the value of tension becomes more negative, the on-rate approaches zero. As discussed, a negative value of tension means the vertex is displaced further out, so it would not be likely for the protrusive force to transition on since it likely transitioned recently into the off state. As tension becomes more positive, the on-rate increases (Fig. 3.7). This is because a positive value of tension means the neighboring vertices are likely protruding so the vertex is likely to transition to an on state to relieve the building up tension and protrude. Biologically, this is analogous to Rac1 signaling, which is responsible for driving protrusion and the formation of lamellipodia. If Rac1 is active at one part of the cell membrane, it is likely to be active nearby to drive protrusion formation and establish polarity. Here, k^*_{on} and T_{max} are constants. k^*_{on} is the base on rate, meaning this is the on rate when the vertex is not under any tension. In the figure below (Fig.3.8), we can see how adjusting k^*_{on} affects k_{on}, where increasing k^*_{on} increases the overall on rate.

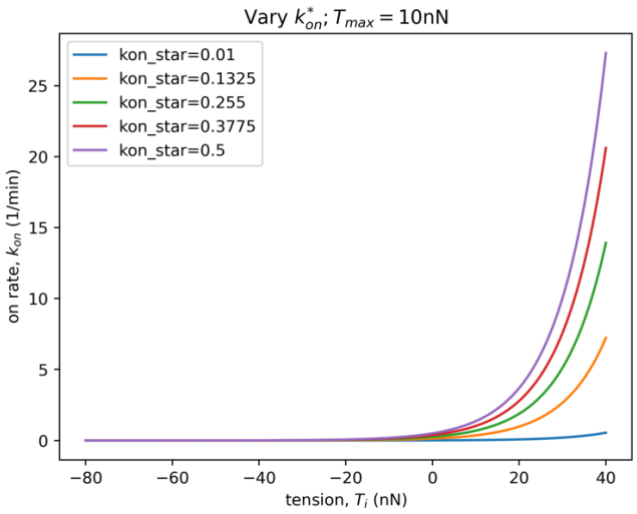


Figure 3.8. On rate for stochastic protrusive force varying k^*_{on}.

Example of how the on rate for the stochastic protrusive force, F_{pr}, varies with tension, T, while varying values of k^*_{on} and maintaining $T_{max} = 10\ nN$.

T_{max} scales the on rate and affects how sensitive the on rates are to tension. In the figure below (Fig. 3.9), we can see that increasing T_{max} decreases the effect of tension on the on rate.

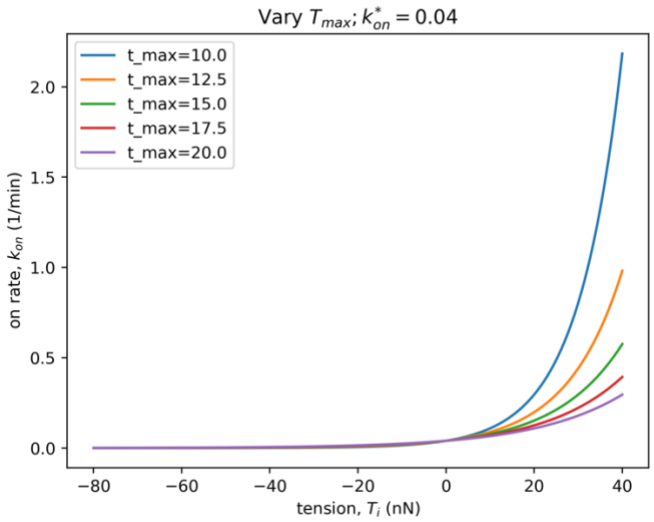


Figure 3.9. On rate for stochastic protrusive force varying T_{max}.

Example of how the on rate for the stochastic protrusive force, F_{pr}, varies with tension, T, while varying values of T_{max} and maintaining $k_{on}^{*} = 0.04\ min^{-1}$.

We can change both k_{on}^{*} and T_{max} to change the behavior of the model. For instance, by decreasing k_{on}^{*}, we decrease overall on rates – a lower value means the cell is not very protrusive regardless of tension and it takes a while for a protrusion to form. By increasing k_{on}^{*}, we increase overall on rates – a higher value means the cell is very protrusive regardless of tension and protrusions can rapidly form but are not necessarily sustained. We can also decrease T_{max} to increase sensitivity of tension at vertex – lower values of T_{max} means a lower tension is required to increase the on rate at a vertex so now vertices are more sensitive to the behavior of their neighbors, thus representing higher "communication" amongst vertices. Conversely, when we Increase T_{max}, we decrease sensitivity of tension at vertex – higher values of T_{max} means a higher tension is required to increase the on rate at a vertex so now vertices are less sensitive to the behavior of their neighbors, now representing less "communication" amongst vertices. It is worth noting that in both cases of changing T_{max}, communication can also be increased or decreased by changing the membrane elasticity, λ, between vertices where a stiffer spring constant would increase communication and vice versa, however, this is not currently being implemented.

Off Rate, k_{off}

The rate the stochastic protrusive force turns off for each vertex is affected by a parameter representing internal cell cytoskeletal polarity, which we will refer to as the internal cell orientation, or simply ω. The internal cell orientation points to a location on the cell perimeter to represent the direction in which the cell is polarized, we will discuss how it updates over time in the next section. The internal cell orientation can point towards any position on the

cell boundary and is not limited to the positions of the vertices. The direction opposite on the cell boundary to where internal cell orientation is pointing is referred to as $\rho_{\omega_{opp}}$ and the position of vertex i along the cell perimeter is ρ_i, as shown in the figure below (Fig. 3.10).

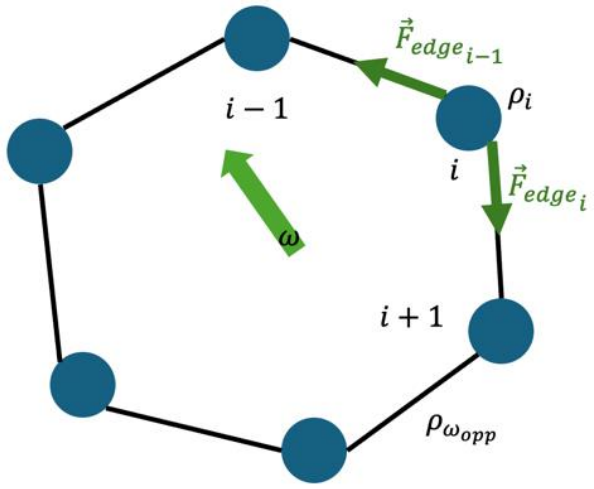


Figure 3.10. Internal cell orientation.

There is an internal cell orientation, ω, that aligns with the direction along the boundary that has the highest tension. The direction antiparallel to this direction is denoted as $\rho_{\omega_{opp}}$. A location of a vertex along the cell boundary is denoted as ρ_i, but a position along the cell boundary, ρ, is continuous and not limited to vertex locations.

The off rate, k_{off}, is given by the following equation:

$$k_{off_i}(\rho_i) = k^*_{off} + k_{off_{max}} e^{-\left(\frac{\rho_i - \rho_{\omega_{opp}}}{\sigma_{off}}\right)^2}$$

Where ρ_i is the continuous position along the cell boundary and $\rho_{\omega_{opp}}$ is the position along the cell boundary opposite to ρ_ω, the position along the boundary in which the internal cell orientation is positioned.

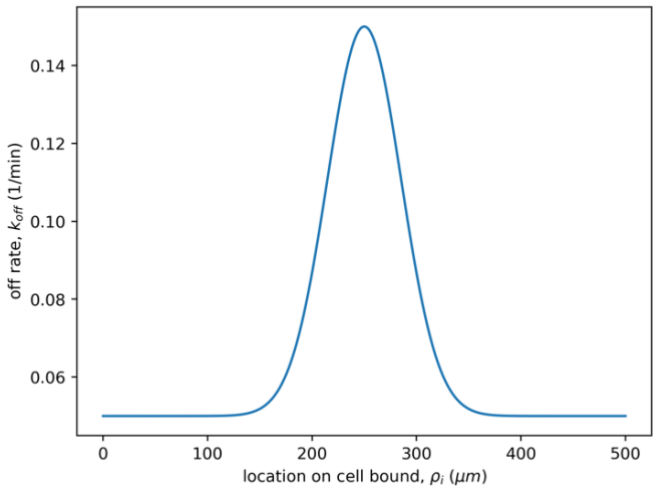


Figure 3.11. Off rate for stochastic protrusive force.

Example of how the off rate for the stochastic protrusive force, F_{pr}, varies with along the cell boundary, ρ_i, using values of $k_{off}^* = 0.05\ min^{-1}$, $k_{off_{max}} = 0.1\ min^{-1}$, and $\sigma_{off} = 30\ \mu m$.

With this form, the off rate is highest for vertices that are directly opposite to ω, and decay exponentially on either side (Fig. 3.11). This helps maintain polarity and cohesiveness in protrusion orientation. Here, k_{off}^*, $k_{off_{max}}$ and σ_{off} are constants. k_{off}^* is the base off rate, meaning this is the rate the stochastic protrusive force turns off when the vertex is not affected by the internal cell orientation. In the figure below (Fig. 3.12) we can see how changing k_{off}^* changes the off rate, where increasing k_{off}^* increases the overall off rate around the entire perimeter of the cell.

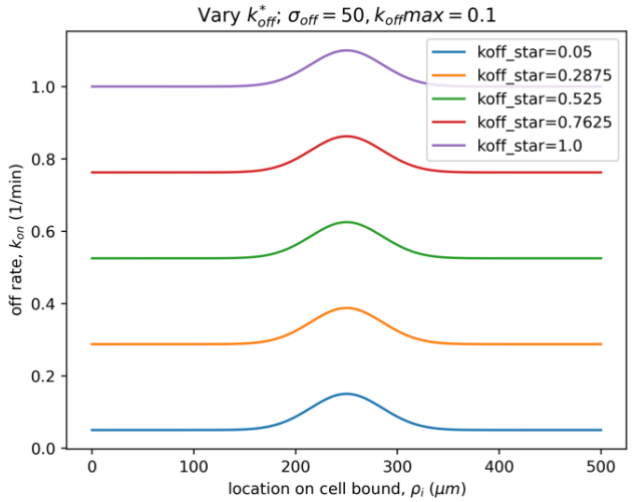


Figure 3.12. Off rate for stochastic protrusive force varying k^*_{off}.

Example of how the off rate for the stochastic protrusive force, F_{pr}, varies with along the cell boundary, ρ_i, varying values of k^*_{off} and maintaining $k_{off_{max}} = 0.1\ min^{-1}$ and $\sigma_{off} = 30\ \mu m$.

$k_{off_{max}}$ is the scaling term for the off rate and affects how sensitive vertices are to the effects of the internal cell orientation. Increasing or decreasing this term will decrease or increase the off rate respectively, as shown below (Fig. 3.13).

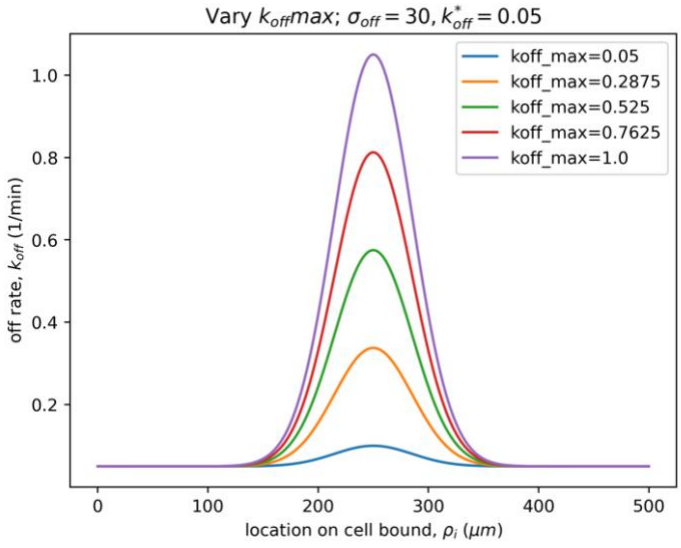


Figure 3.13. Off rate for stochastic protrusive force varying $k_{off_{max}}$.

Example of how the off rate for the stochastic protrusive force, F_{pr}, varies with along the cell boundary, ρ_i, varying values of $k_{off_{max}}$ and maintaining $k^*_{off} = 0.05\ min^{-1}$ and $\sigma_{off} = 30\ \mu m$.

σ_{off} affects the width of the off-rate function and thus determines how many vertices' off rates are affected by the internal cell orientation. As shown in the figure below (Fig. 3.14), by increasing σ_{off} a wider range of vertices near $\rho_{\omega_{opp}}$ have an increased off rate while decreasing σ_{off} localizes the effect of the off rate.

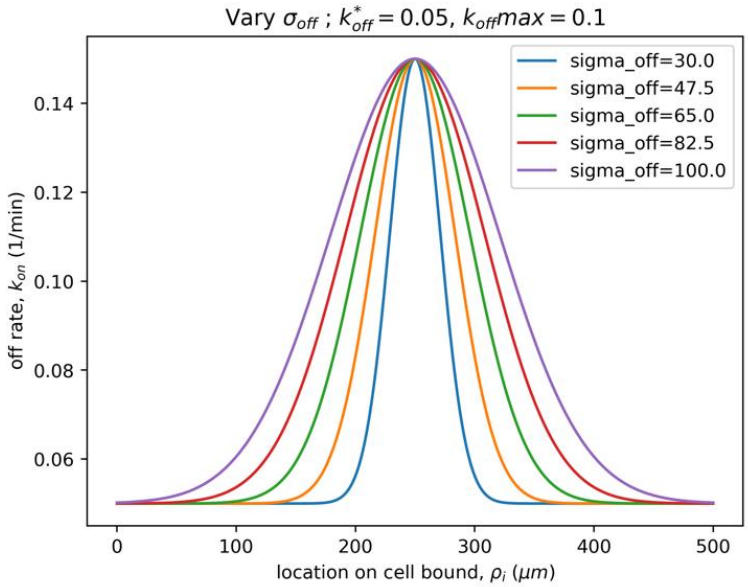


Figure 3.14. Off rate for stochastic protrusive force varying σ_{off}.

Example of how the off rate for the stochastic protrusive force, F_{pr}, varies with along the cell boundary, ρ_i, varying values of σ_{off} and maintaining $k^*_{off} = 0.05\ min^{-1}$ and $k_{off_{max}} = 0.1\ min^{-1}$.

To summarize, we can change k^*_{off}, $k_{off_{max}}$ and σ_{off} to change the behavior of the model. We can increase k^*_{off} to increase overall off rate around cell perimeter and decrease k^*_{off} to decrease

overall off rate around cell perimeter. We can increase $k_{off_{max}}$ to make the internal cell orientation have a larger effect on the off rate of relevant vertices or decrease $k_{off_{max}}$ to decrease the effect of the internal cell orientation of relevant vertices. Lastly, we can increase σ_{off} to increase the number of vertices that are affected by the internal cell orientation or decrease σ_{off} to decrease and localize the effect of the internal cell orientation.

Updating the Internal Cell Orientation, ω

The internal cell orientation, ω, points to the location on the cell boundary under greatest net tension. Once again, tension is defined as:

$$T_i = \left(\vec{F}_{edge_{i-1}} + \vec{F}_{edge_i}\right) \cdot \vec{n}_i$$

And net tension is defined as:

$$\vec{T}_{net} = \sum_{i=1}^{N} T_i \vec{n}_i$$

Where N is the number of vertices that make up the cell perimeter. The internal cell orientation updates according to the equation:

$$\eta_\omega \frac{d\theta}{dt} = F_{align}(\vec{r}_i)\theta + \tilde{g}(t)$$

Where:

$$F_{align}(\vec{r}_i) = \frac{\left|\vec{T}_{net}\right|/N}{T_{avg}}$$

And $T_{avg} = (1/N)\sum_{i=1}^{N} T_i$.

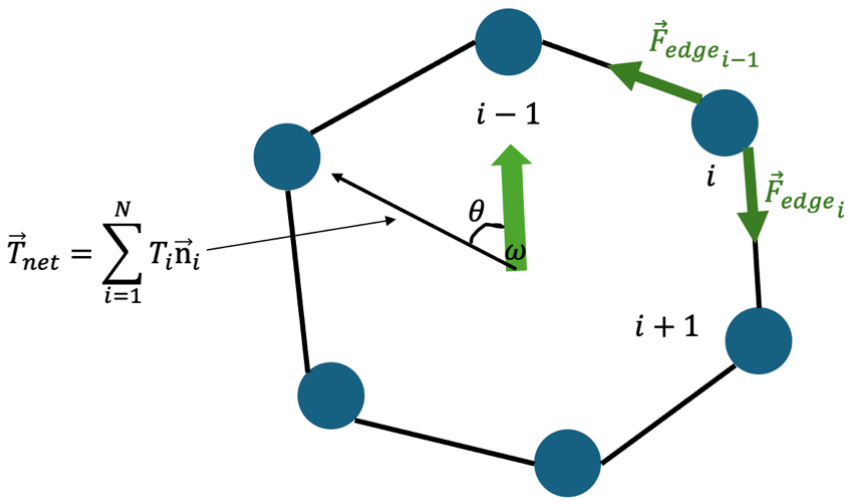


Figure 3.15. Updating internal cell orientation.
The internal cell orientation, ω, updates such that it aligns with the point along the boundary that has the highest net tension, $\vec{T}_{net}$.

Here, θ is the angle between the internal cell polarity and the direction of net tension on the cell border. $F_{align}(\vec{r}_i)$ is the alignment force and acts such that the greater the net tension relative to the average of tensions at each vertex the greater the force (Fig. 3.15). η_ω is the friction coefficient for the internal cell orientation and it's time scale is related to the time necessary for the cell to disassemble and reassemble its cytoskeleton in a new orientation to adapt to the change in polarity. Lastly, $\tilde{g}(t)$ is a Gaussian white noise process where $\tilde{g}(t) = \sigma_g \eta(t)$. Here, σ_g is the strength of the process and $\eta(t)$ is Gaussian white noise.

While we have implemented F_{align} as depending on the magnitude of the tension relative to the average tension around the cell, it may be sufficient for this force to be a constant. Regardless, the value of η_ω could be fit by looking at data of cells migrating and using the major force dipole axis from traction force data to determine polarization and see how this dipole changes direction over time.

Updating the Positions of Vertices, $\vec{r}_i$

The positions of each vertex are updated using the following equation:

$$\eta_{v_i} \frac{d\vec{r}_i}{dt} = \vec{F}_{edge_{i-1}} + \vec{F}_{edge_i} + \left(p_i + F_{pr_i}(t)\right)\vec{n}_i + \tilde{f}(t)$$

We have already defined $\vec{F}_{edge_i}$, $\vec{p}_i$, and $\vec{F}_{pr_i}(t)$ in the sections above. η_{v_i} is the friction coefficient for each vertex and we will discuss how it is determined in the next section. Lastly, $\tilde{f}(t)$ is a Gaussian white noise process where $\tilde{f}(t) = \sigma_f \eta(t)$. Here, σ_f is the strength of the process and $\eta(t)$ is Gaussian white noise.

Friction Coefficient of Vertices, η_{v_i}

The vertices can be in two different states: adhesive and unstuck, which we will denote as:

$$H_{v_i}(t) = \begin{cases} \eta_{adhesive} \\ \eta_{unstuck} \end{cases}$$

If $H_{v_i}(t) = \eta_{adhesive}$, then that means that the vertex can be "adhered" to the surface, thus yielding a higher friction coefficient. Vertices that are directly opposite to the internal cell orientation have a higher friction coefficient. If the distance of a vertex to the center of the cell, d_{v_i}, is greater than some maximum value, d_{max}, then the state of the vertex becomes $H_{v_i}(t) = \eta_{unstuck}$, meaning it is no longer able to be "adhered" to surface and instead has some constant value for the friction coefficient. The function for the friction coefficient is given below:

$$\eta_{v_i}(t) = \begin{cases} \eta_v^* + \eta_{max} e^{-\left(\frac{\rho_i - \rho_{\omega_{opp}}}{\sigma_\eta}\right)^2}, & H_{v_i}(t) = \eta_{adhesive} \\ \eta_v^*, & H_{v_i}(t) = \eta_{unstuck} \end{cases}$$

Once again, the direction opposite on the cell boundary to where internal cell orientation is pointing is referred to as $\rho_{\omega_{opp}}$ and the position of vertex i along the cell perimeter is ρ_i. Here, η_v^*, η_{max}, and σ_η are constants. η_v^* is the base friction coefficient for each vertex. η_{max} is the scaling term for friction coefficient and affects how sensitive vertices are to the effects of the internal cell orientation. σ_η determines the width of the function, so it affects how many vertices are affected by the internal cell orientation. Once in the $\eta_{unstuck}$ state, the vertex can renter the $\eta_{adhesive}$ state with some rate, $k_{adhesive_i}$. This is illustrated in the diagram below (Fig. 3.16):

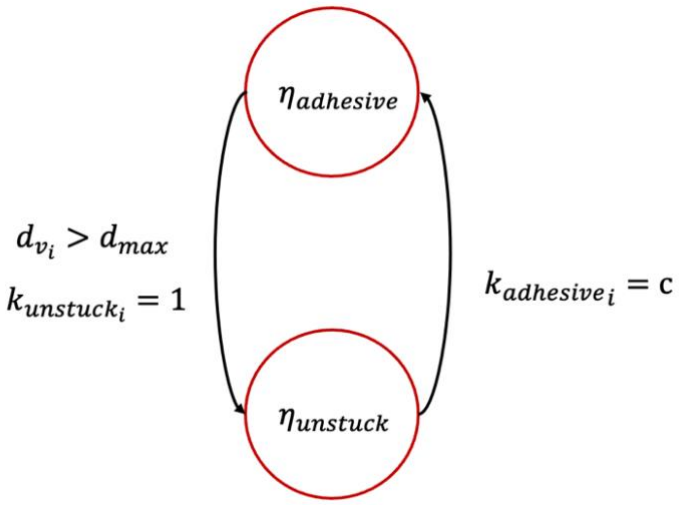


Figure 3.16. Adhesion states at each vertex.

Each vertex can be in a stuck or unstuck state, where transitions to the adhesive state occur at a constant rate, $k_{adhesive}$, and transitions to the unstuck state occur if the distance of a vertex to the center of the cell, d_{v_i}, is greater than some maximum value, d_{max}.

There are several parameters to change and fit here: η_v^* for base unadhered friction coefficient of vertex, η_{max} for the maximum values of friction coefficient directly opposite to internal cell orientation (unreasonably large values where the vertex can't move with given forces are equivalent to being adhered), σ_η for how many vertices' friction coefficients are affected by the internal cell orientation, d_{max}, the maximum distance a vertex can be from the center of the cell before it becomes unattached from the surface, and, lastly, $k_{adhesive_i}$, the rate a vertex can become adhesive again once it becomes unattached. It is worth noting that if a vertex is opposite the internal cell orientation, it has a higher friction coefficient and is therefore adhered to the surface to some degree. However, as the model currently stands, if d_i for the vertex that is adhered never exceeds d_{max}, it technically never becomes unattached and instead as the internal cell orientation shifts away from that vertex the friction coefficient gradually becomes smaller. This opens the question if entering the unattached state should instead be determined by magnitude of the friction coefficient where if $\eta_{v_i} \geq \eta_{max}$ so the vertex becomes unadhesive.

Model Assumptions

There are several assumptions made by current model along with proposed mechanisms. Firstly, the on rate for protrusive force, k_{on}, depends on tension along cell edge where higher outward tension means a higher on rate. Next, the off rate for protrusive force, k_{off}, depends on the internal cell orientation where off rate is highest directly opposite to cell direction of polarization (internal cell orientation). Additionally, this direction of polarization depends on tension along the cell edge where higher inward tension means a higher direction of polarization. Lastly, friction coefficients of vertices (cell edge), η_{v_i}, depends on internal cell orientation where friction is highest directly opposite to cell direction of polarization, so a high off rate means a

high friction coefficient. Vertices are sites where focal adhesions form and these adhesions become unstuck from the surface when their distance to the center of the cell exceeds some maximal distance.

3.4 Results

Numerically solving the ODEs for vertex positions and internal cell orientation using Euler's method (Appendix), we can run the simulation forward in time for our simulated cell (Fig. 3.17). Over time, the simulated cell changes shape and has a net displacement.

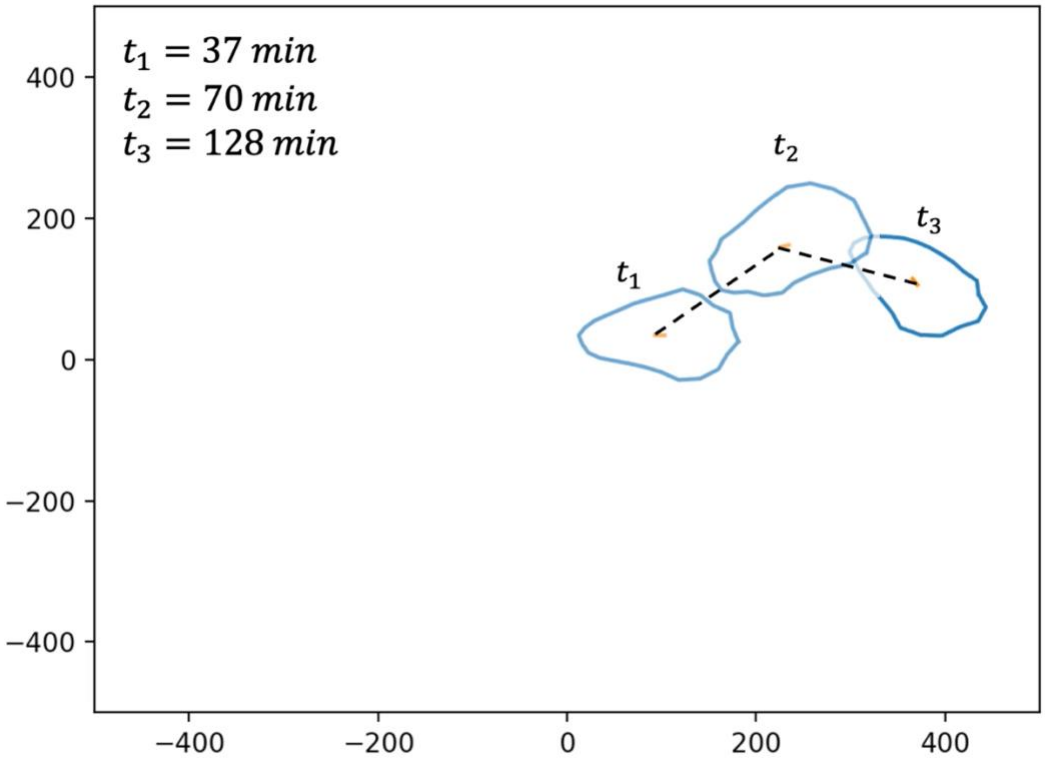


Figure 3.17. Simulating the VBM through time.

Snapshots of a VBM simulation at three different time points.

Given this model is stochastic, it is more insightful to gain insights from a population rather than a single cell. We can repeat this process many different times to simulate a population of cells to observe the effect of changing various parameters within the model.

Changing the On Rate of Protrusion

First, we will explore how changing the on rate of protrusion at each vertex effects the behavior of the model. One way to increase the on rate, k_{on}, is to increase the base on rate, k_{on}^* (Fig. 3.18).

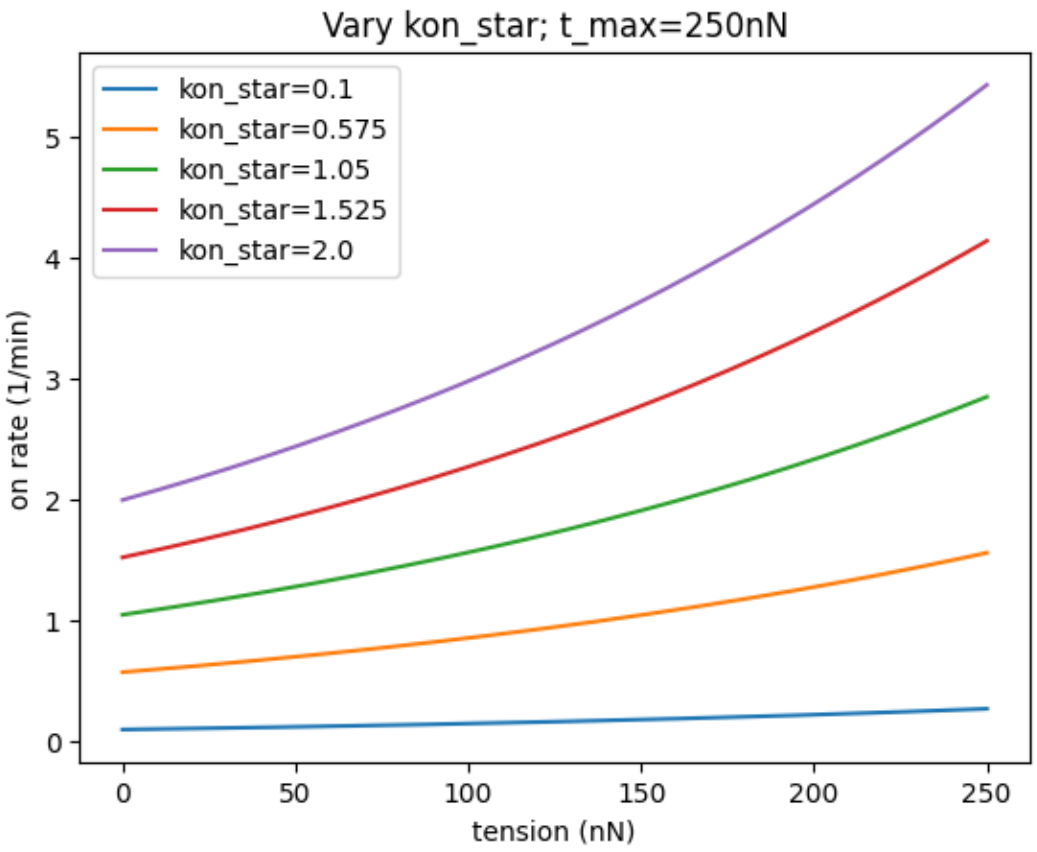


Figure 3.18. Varying k_{on}^* to change the protrusive on rate, k_{on}.
Effect of varying k_{on}^* to change the protrusive on rate.

When we do this with a population of 10 cells, we can see that increasing k^*_{on} decreases cell speed and persistence (Fig. 3.19).

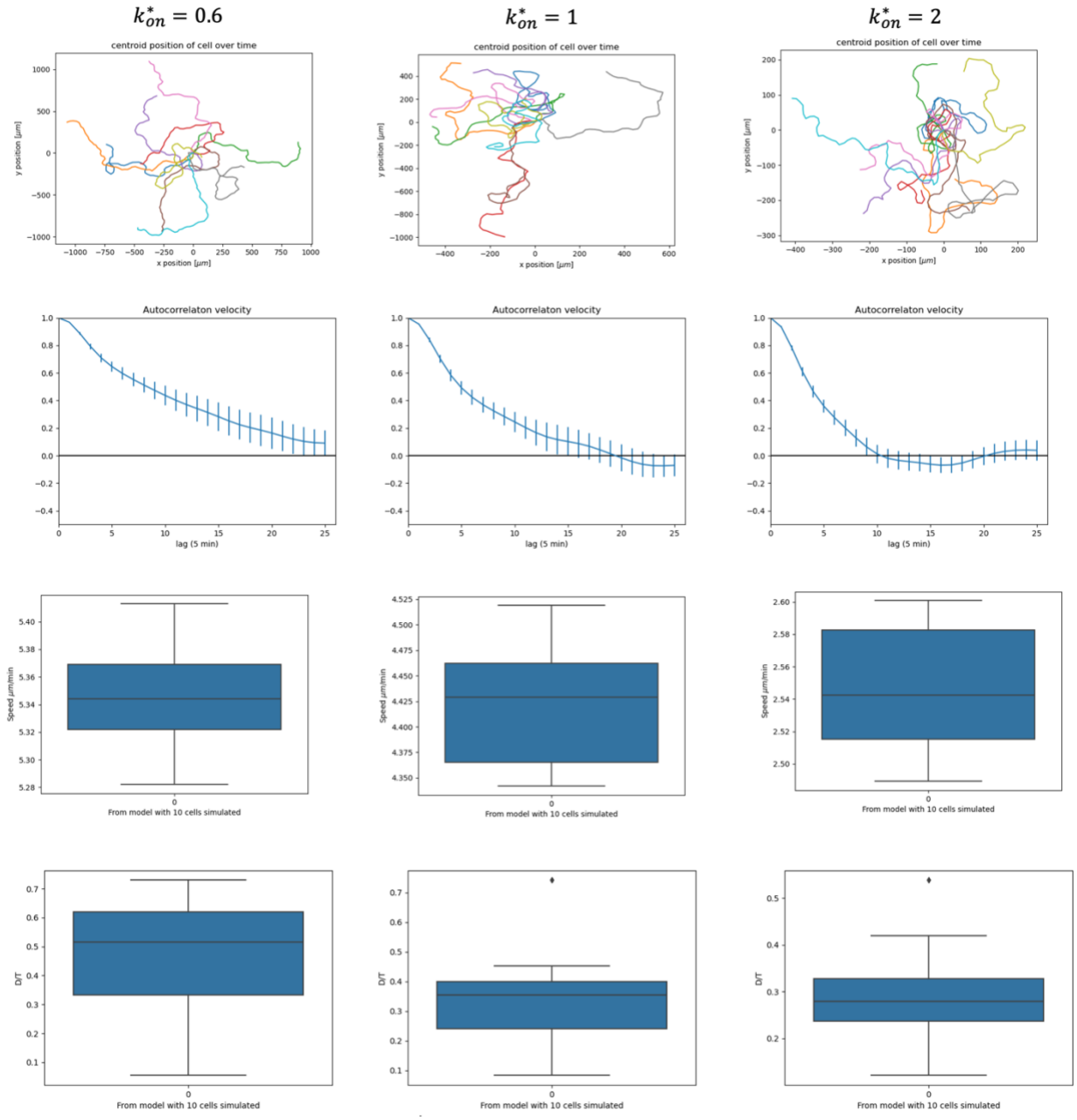


Figure 3.19. Effect of varying k^*_{on} on a population of cells.

A simulation of 10 cells with the following parameters: $T_{max} = 250$, $k^*_{off} = 0.05$, $k_{off_{max}} = 5$, $\sigma_{off} = 100$.

Another way to increase k_{on} is to decrease T_{max}, which affects how sensitive the on rates are to tension between vertices (Fig. 3.20).

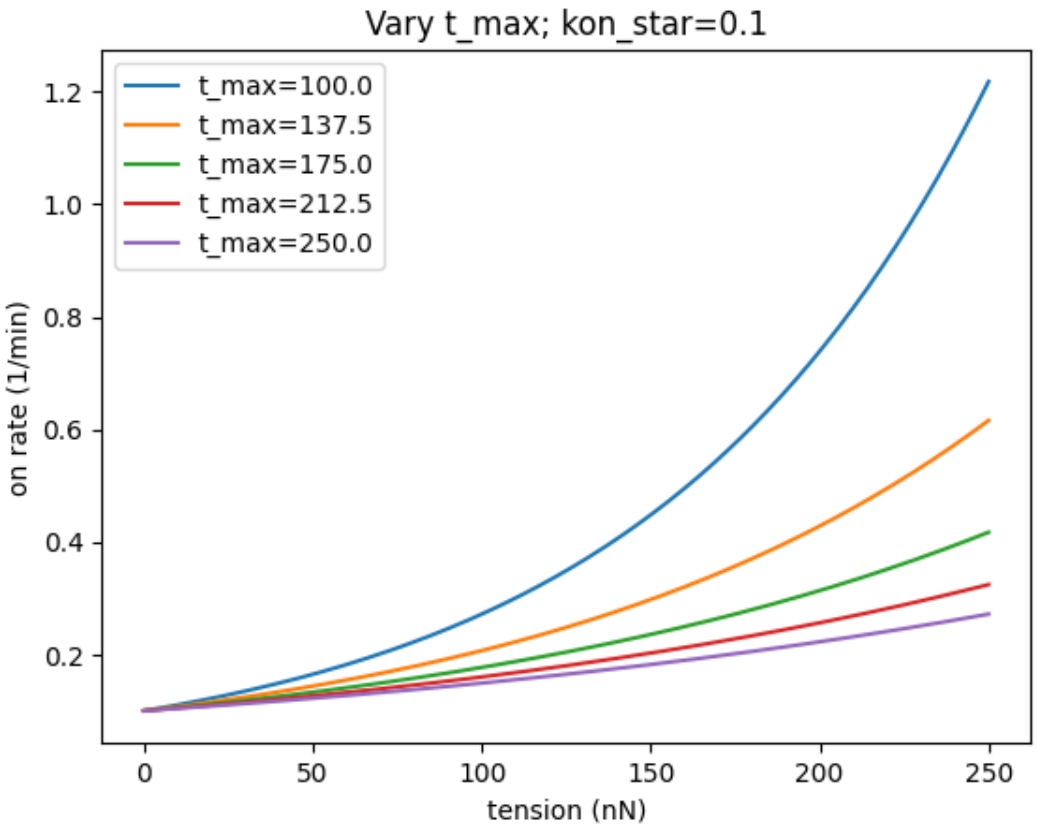


Figure 3.20. Varying T_{max} to change the protrusive on rate, k_{on}.
Effect of varying T_{max} to change the protrusive on rate.

We can see that by increasing T_{max}, we appear to decrease cell speed and increase persistence (Fig. 3.21).

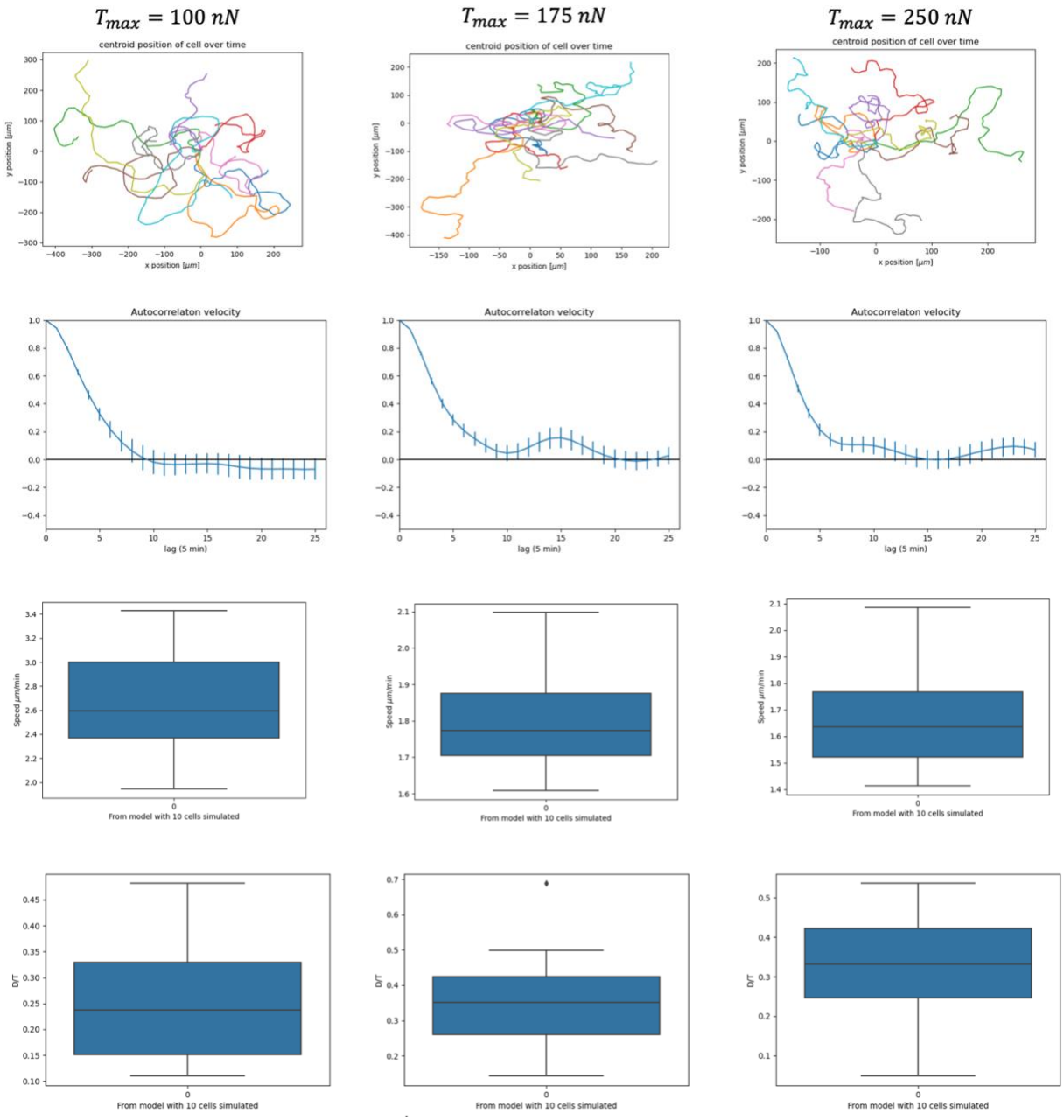


Figure 3.21. Effect of varying T_{max} on a population of cells.

A simulation of 10 cells with the following parameters: $k^*_{on} = 0.1$, $k^*_{off} = 0.05$, $k_{off_{max}} = 5$, $\sigma_{off} = 100$.

Lastly, we can vary both k^*_{on} and T_{max} to change the on rate (Fig. 3.22).

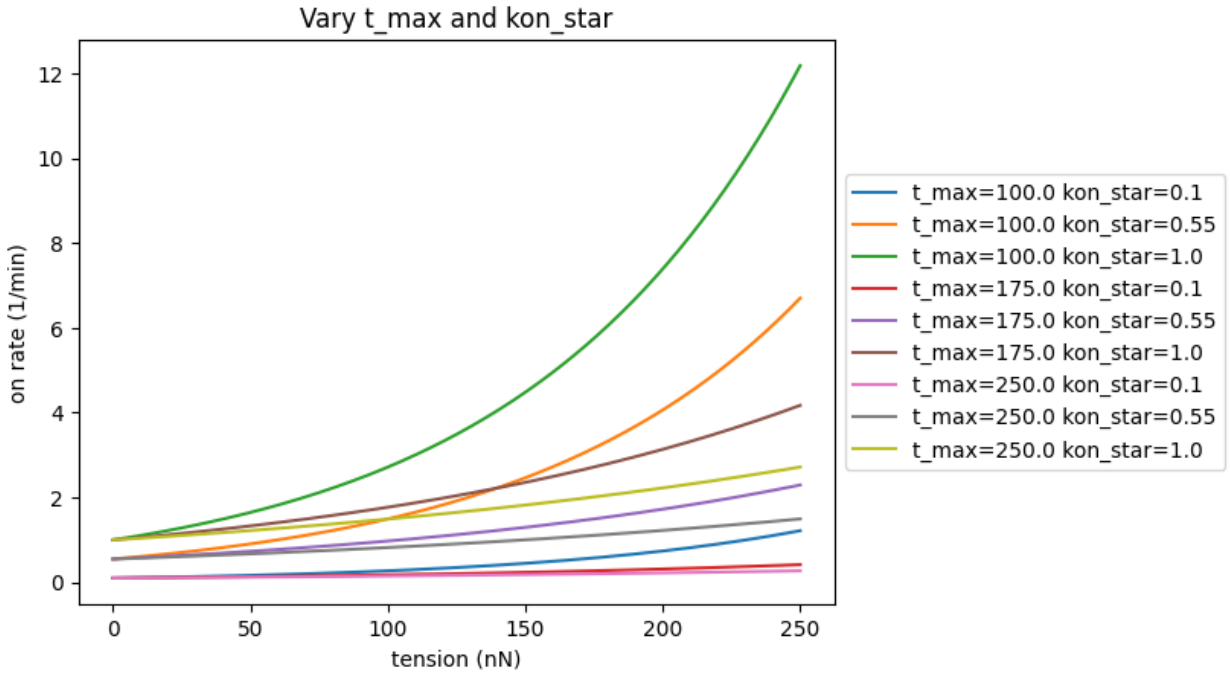


Figure 3.22. Varying k_{on}^{*} and T_{max} to change the protrusive on rate, k_{on}.

Effect of varying k_{on}^{*} and T_{max} to change the protrusive on rate.

By changing both k_{on}^{*} and T_{max} to decrease the on rate, we see that cell speed and persistence increases (Fig. 3.23).

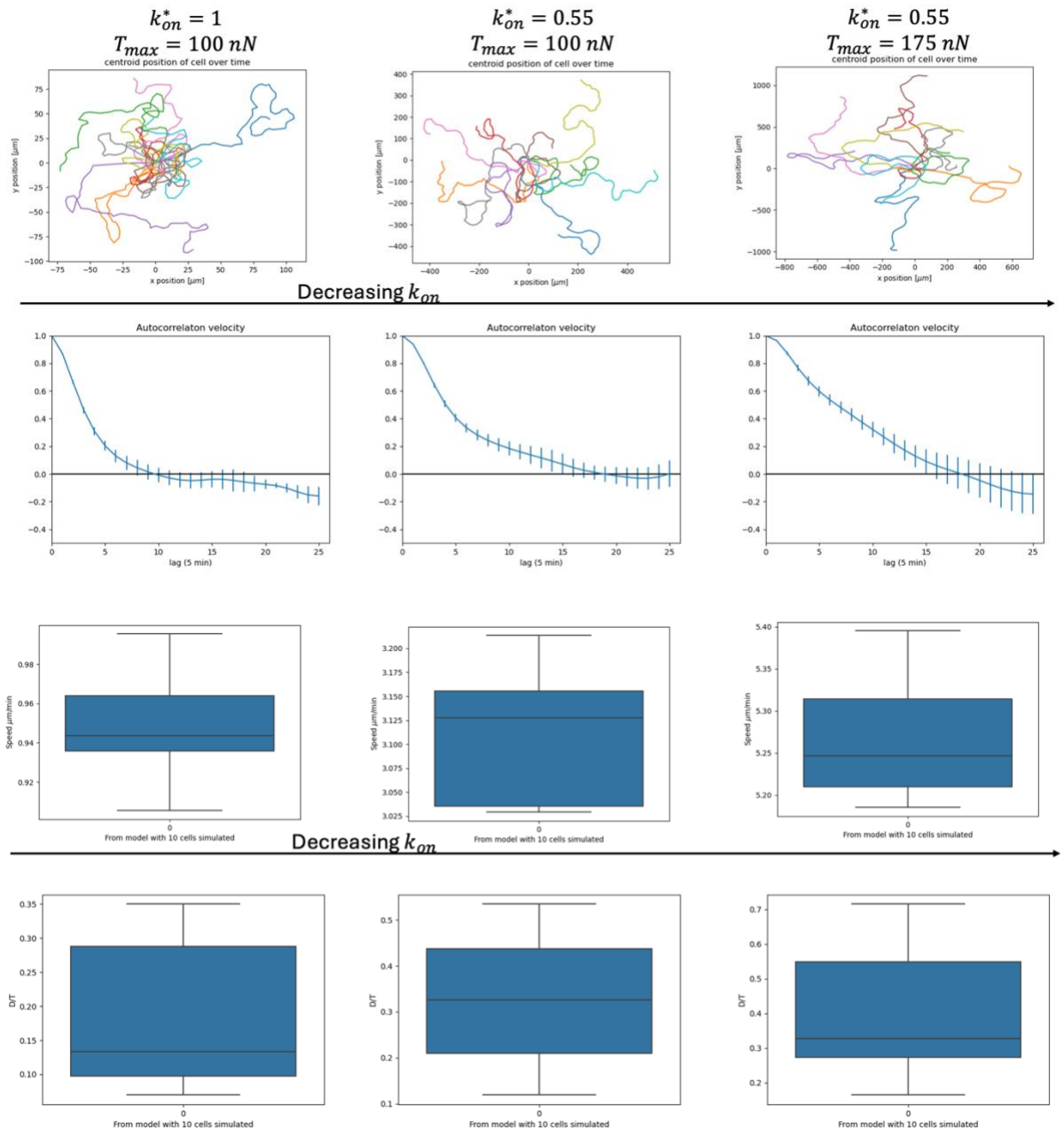


Figure 3.23. Effect of varying k_{on}^* and T_{max} on a population of cells.

A simulation of 10 cells with the following parameters: $k_{off}^* = 0.05$, $k_{off_{max}} = 5$, $\sigma_{off} = 100$.

One thing worth noting is that it appears that changing k_{on}^{*} has a bigger effect on changing persistence than T_{max}. However, overall, increasing k_{on} makes cells slower and less directed in motion.

Changing the Off Rates of Protrusion

Similar to the on rate, there are different ways to decrease the off rates for protrusion. One way to increase the off rate, k_{off}, is to increase the base off rate, k_{off}^{*} (Fig. 3.24). By doing this we also put less importance on how the internal cell orientation affects the off rate as well.

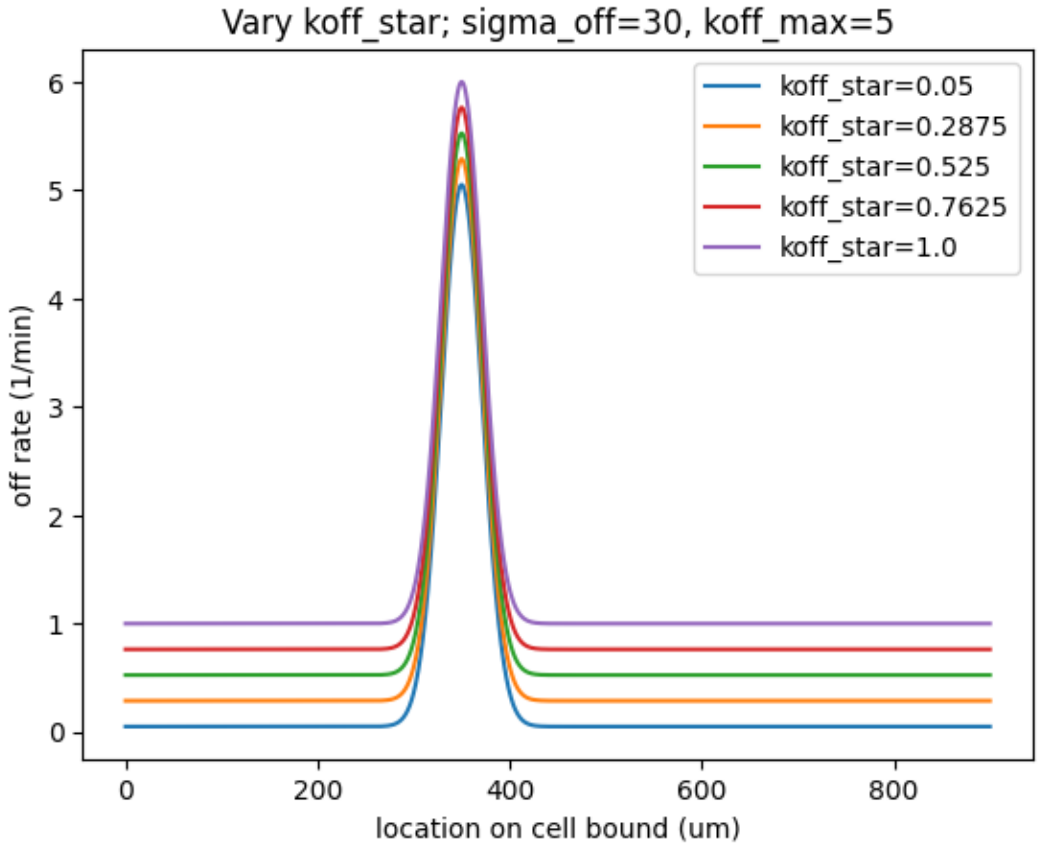


Figure 3.24. Varying k_{off}^{*} to change the protrusive off rate, k_{off}.

Effect of varying k_{off}^{*} to change the protrusive off rate.

By increasing k^*_{off}, speed is decreased without much change to persistence (Fig. 3.25).

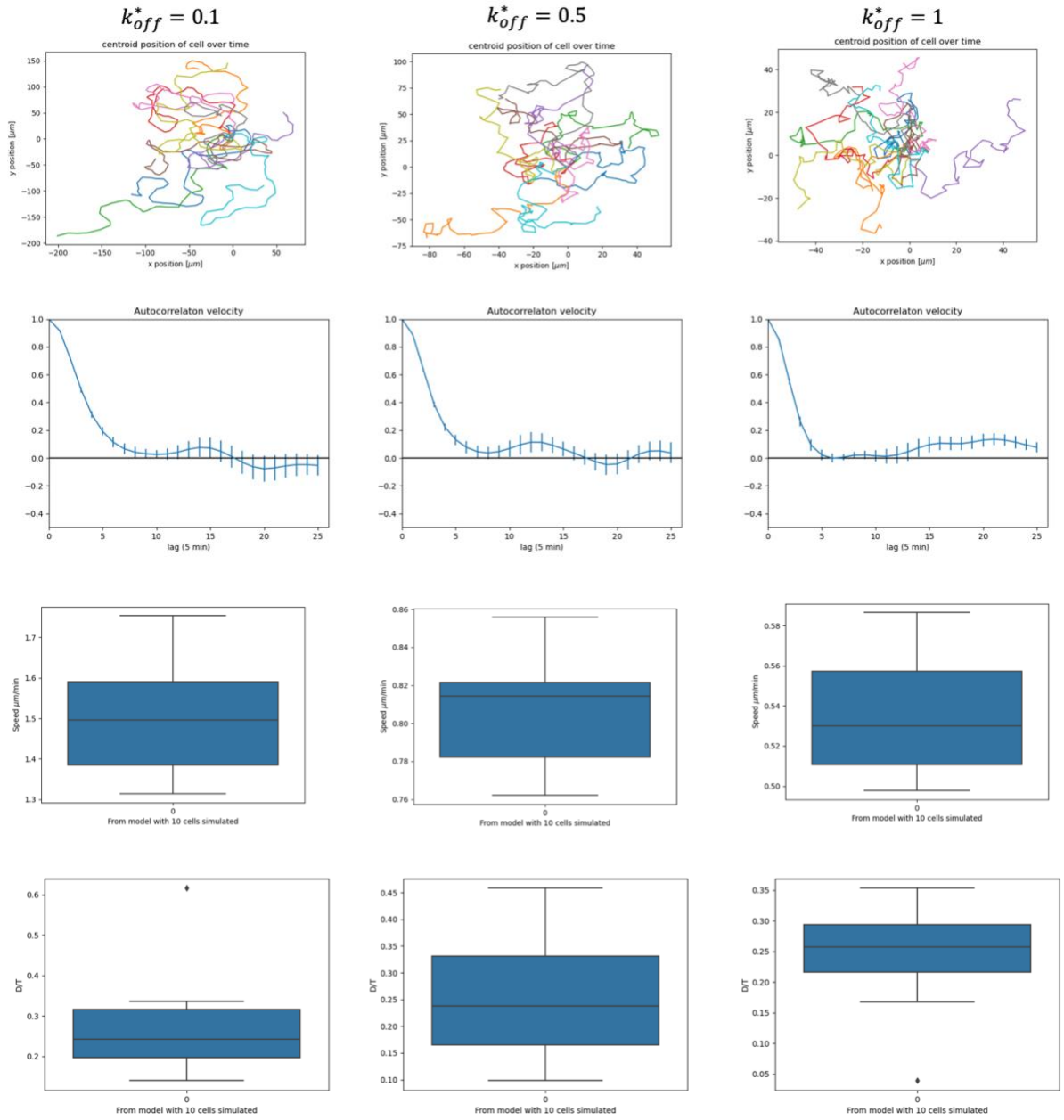


Figure 3.25. Effect of varying k^*_{off} on a population of cells.

A simulation of 10 cells with the following parameters: $k_{off_{max}} = 5$, $\sigma_{off} = 100$.

We can also change the off rate by varying $k_{off_{max}}$, the scaling term for the off rate that also affects how sensitive vertices are to the effects of the internal cell orientation. By increasing $k_{off_{max}}$, the off rate increases for only those vertices that are affected by the internal cell orientation (Fig. 3.26).

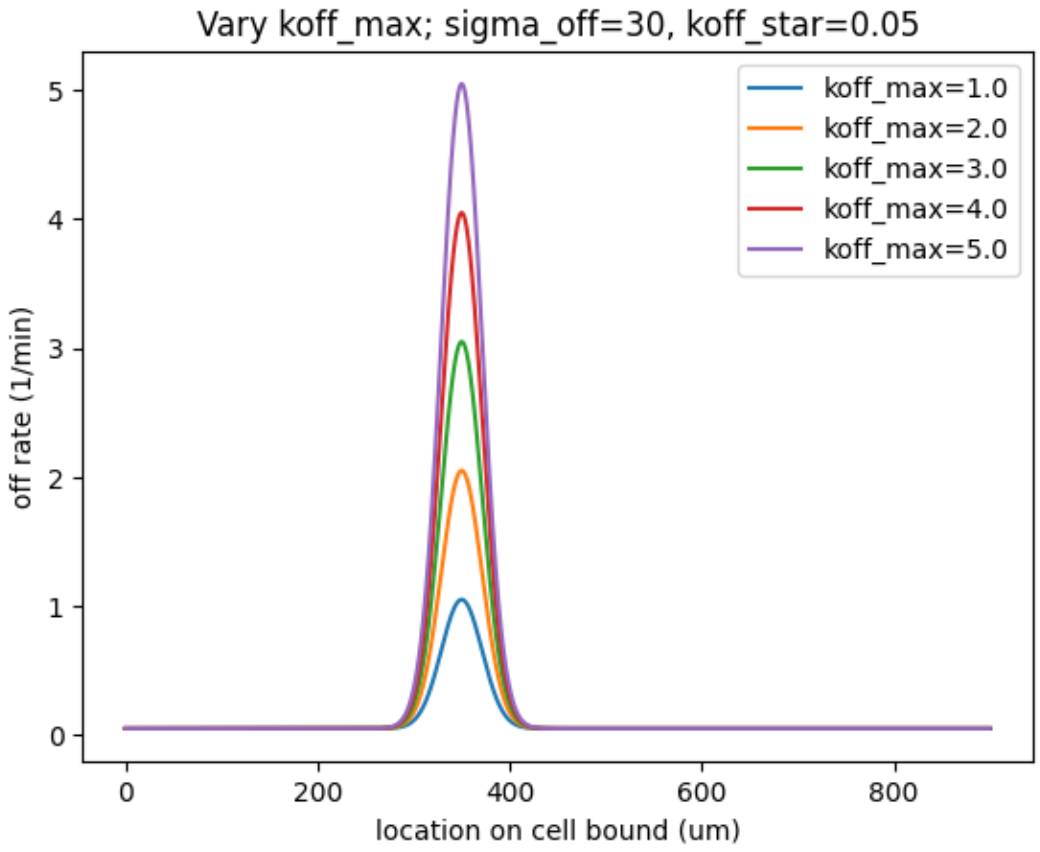


Figure 3.26. Varying $k_{off_{max}}$ to change the protrusive off rate, k_{off}.
Effect of varying $k_{off_{max}}$ to change the protrusive off rate.

Interestingly, it appears there is an optimal value for $k_{off_{max}}$ to maximize cell speed and persistence (Fig. 3.27).

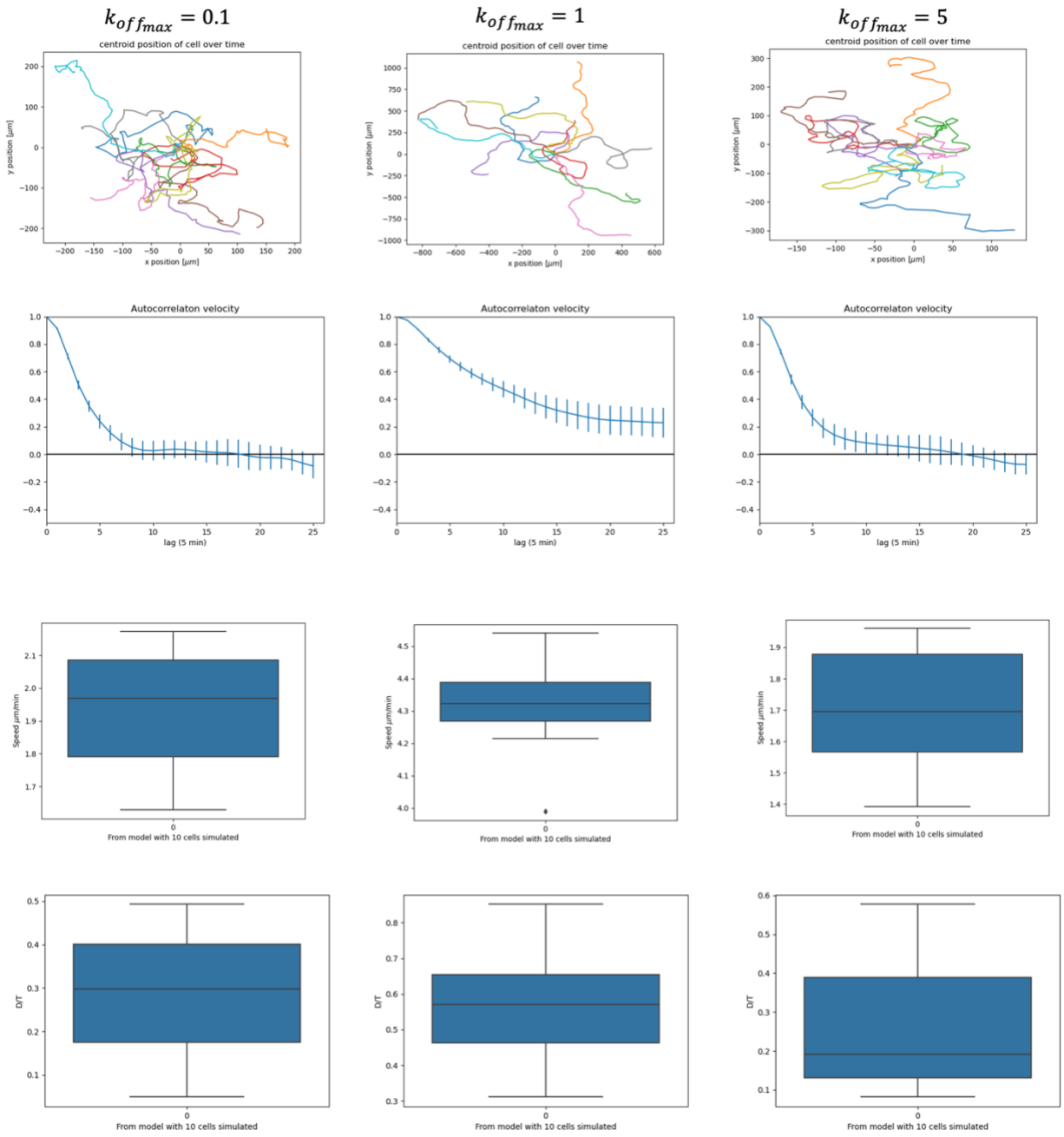


Figure 3.27. Effect of varying $k_{off_{max}}$ on a population of cells.

A simulation of 10 cells with the following parameters: $k^*_{off} = 0.05$, $\sigma_{off} = 100$.

Lastly, we can change the off rate by varying σ_{off}. The effect of increasing σ_{off} increases off rates for a wider range of neighboring vertices (Fig. 3.28).

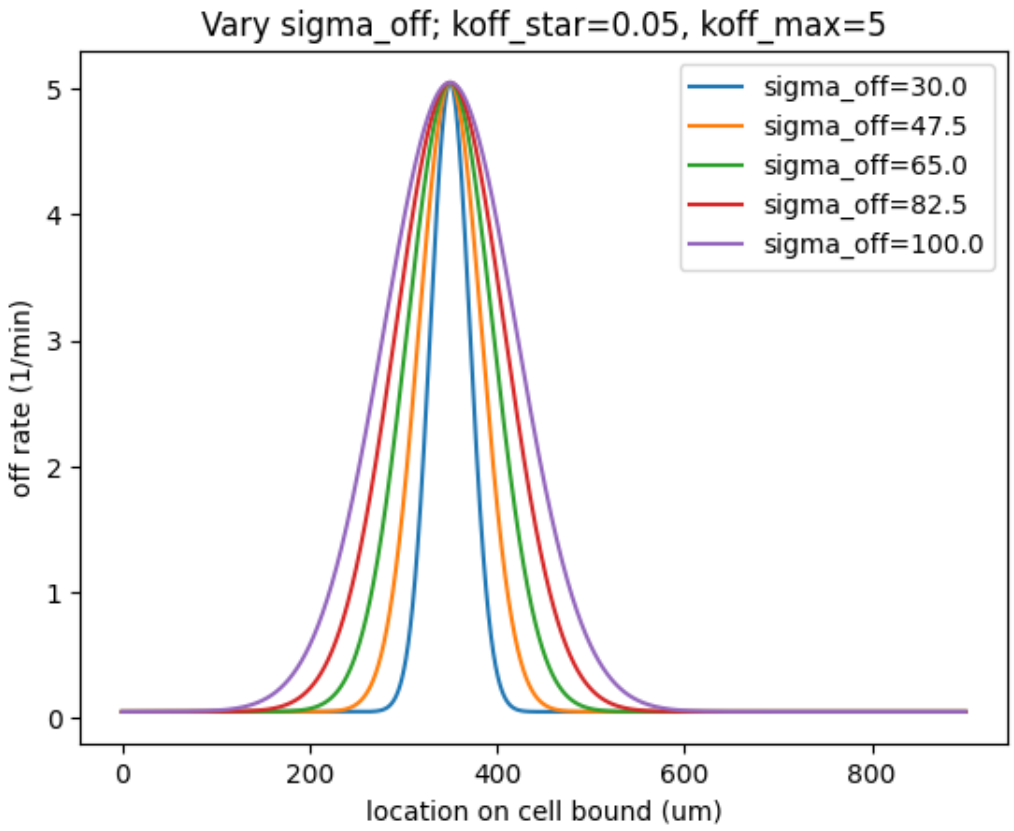


Figure 3.28. Varying σ_{off} to change the protrusive off rate, k_{off}.

Effect of varying σ_{off} to change the protrusive off rate.

Similar to changing $k_{off\,max}$, it appears there is also an optimal value that maximizes cell speed and persistence (Fig. 3.29).

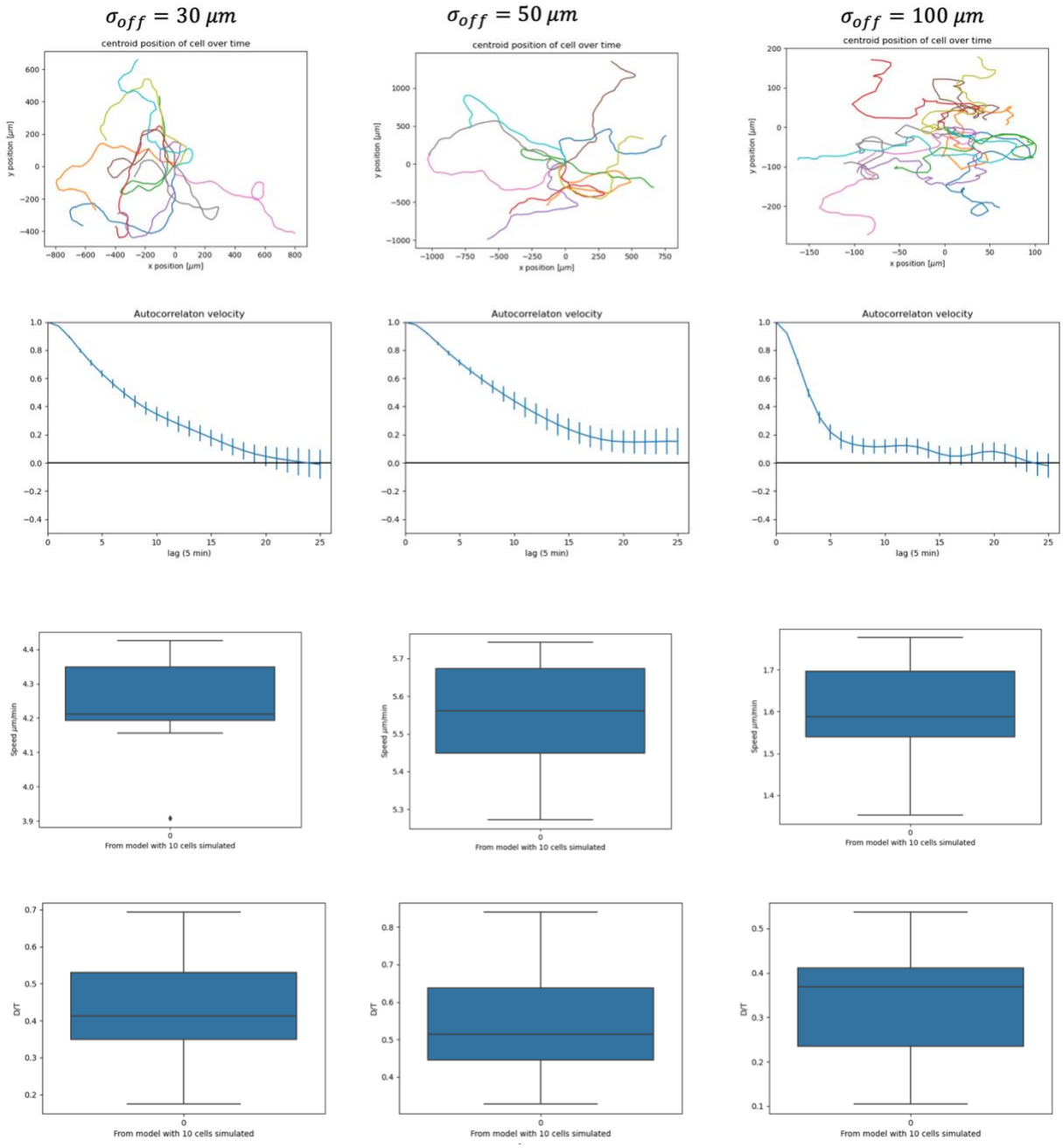


Figure 3.29. Effect of varying σ_{off} on a population of cells.

A simulation of 10 cells with the following parameters: $k^*_{off} = 0.05$, $k_{off\,max} = 5$.

Changing the Friction Coefficients

There are several ways to vary the friction coefficients at each vertex given its form. First, we can increase η_v^* to increase the minimum friction coefficient at each vertex (Fig. 3.30). This has the effect of overall increasing the friction coefficient at each vertex.

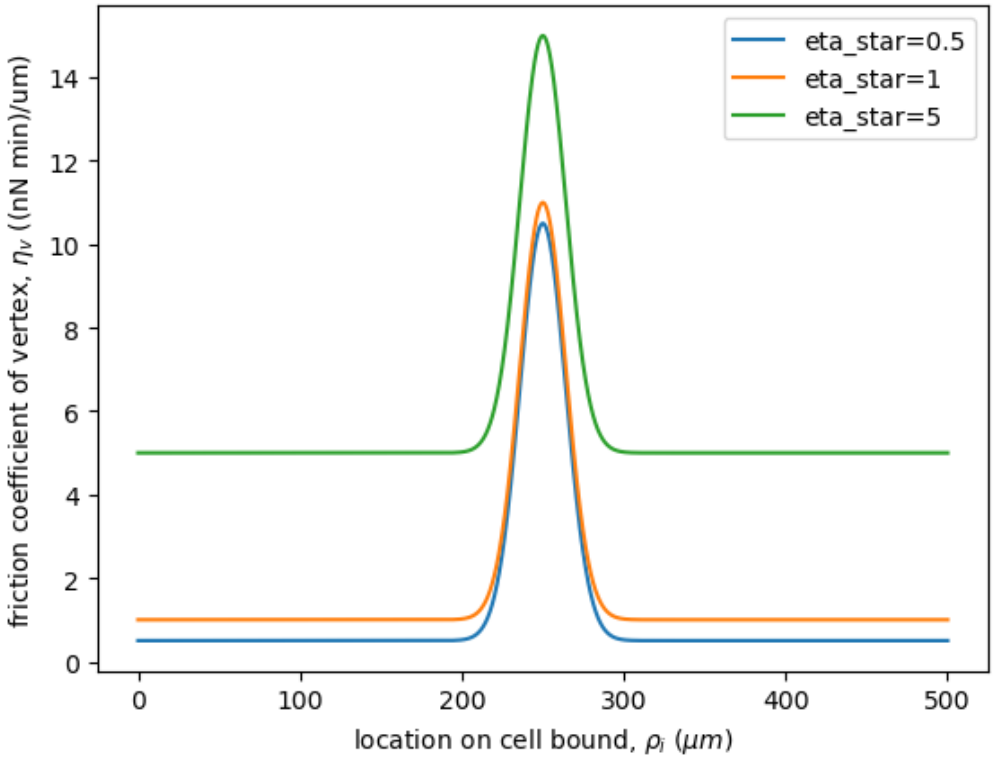


Figure 3.30. Varying η_v^* to change the friction coefficient at each vertex, η_{v_i}.

Effect of varying η_v^* to change the friction coefficients at each vertex.

By increasing the minimum friction value, cell speed decreases and persistence increases (Fig. 3.31).

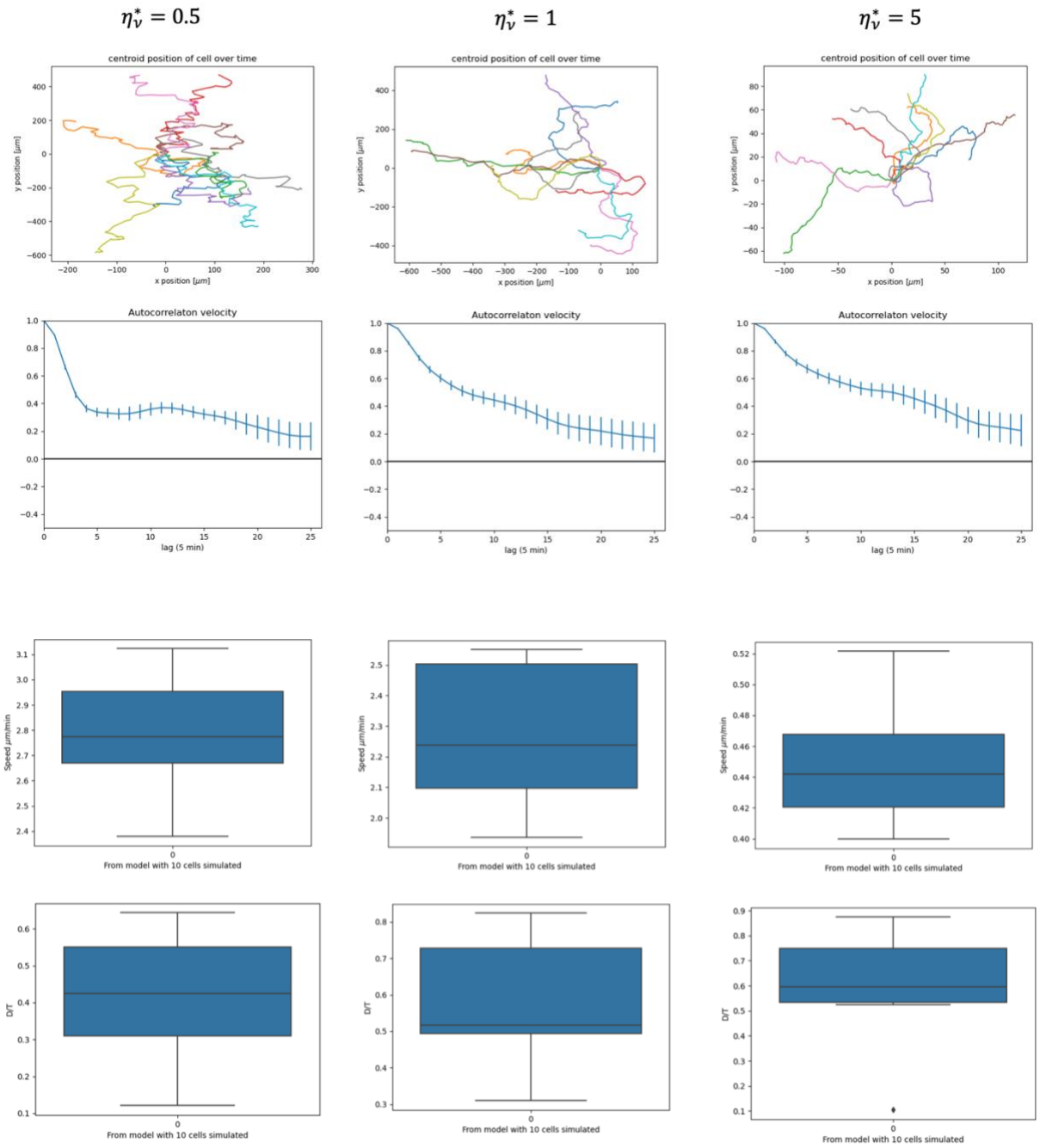


Figure 3.31. Effect of varying η_v^* on a population of cells.

A simulation of 10 cells with the following parameters: $\eta_{max} = 500$, $\sigma_\eta = 20, d_{max} = 60$.

Another way to change the friction coefficient is by increasing η_{max}, which increases the maximum value of the friction coefficient a vertex can experience (Fig. 3.32).

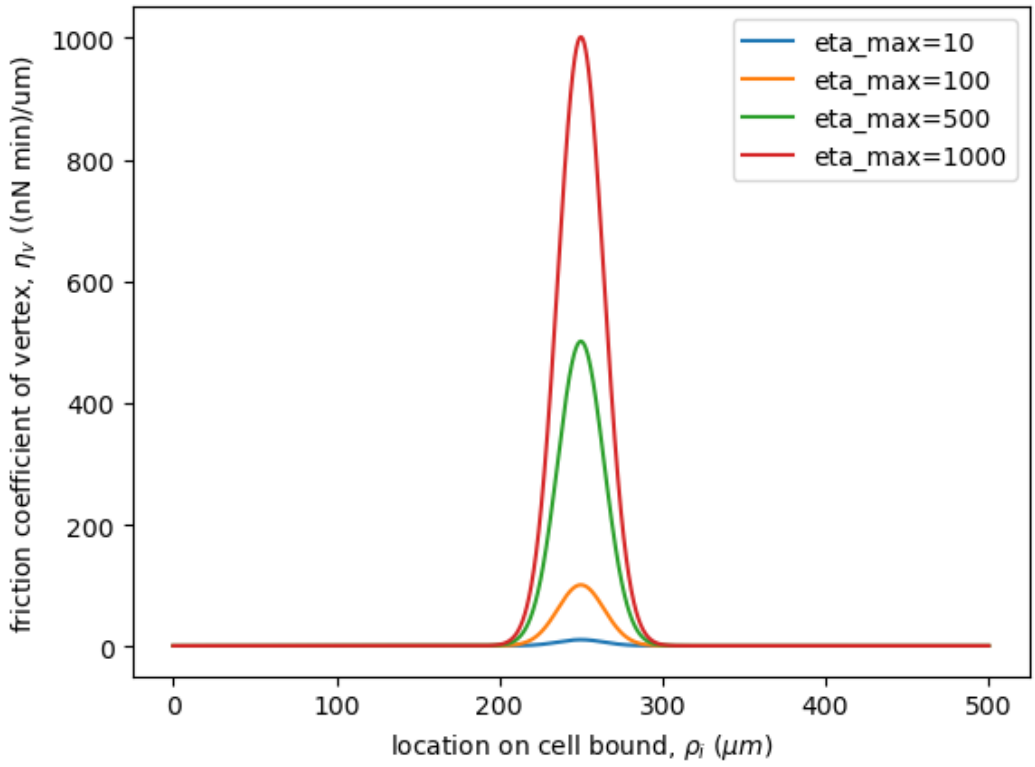


Figure 3.32. Varying η_{max} to change the friction coefficient at each vertex, η_{v_i}.
Effect of varying η_{max} to change the friction coefficients at each vertex.

Increasing the maximum friction value has little effect (Fig. 3.33). This is not surprising since past a certain limit a high friction coefficient is equivalent to being "stuck" to the surface so it cannot be moved by the forces generated by the simulated cell.

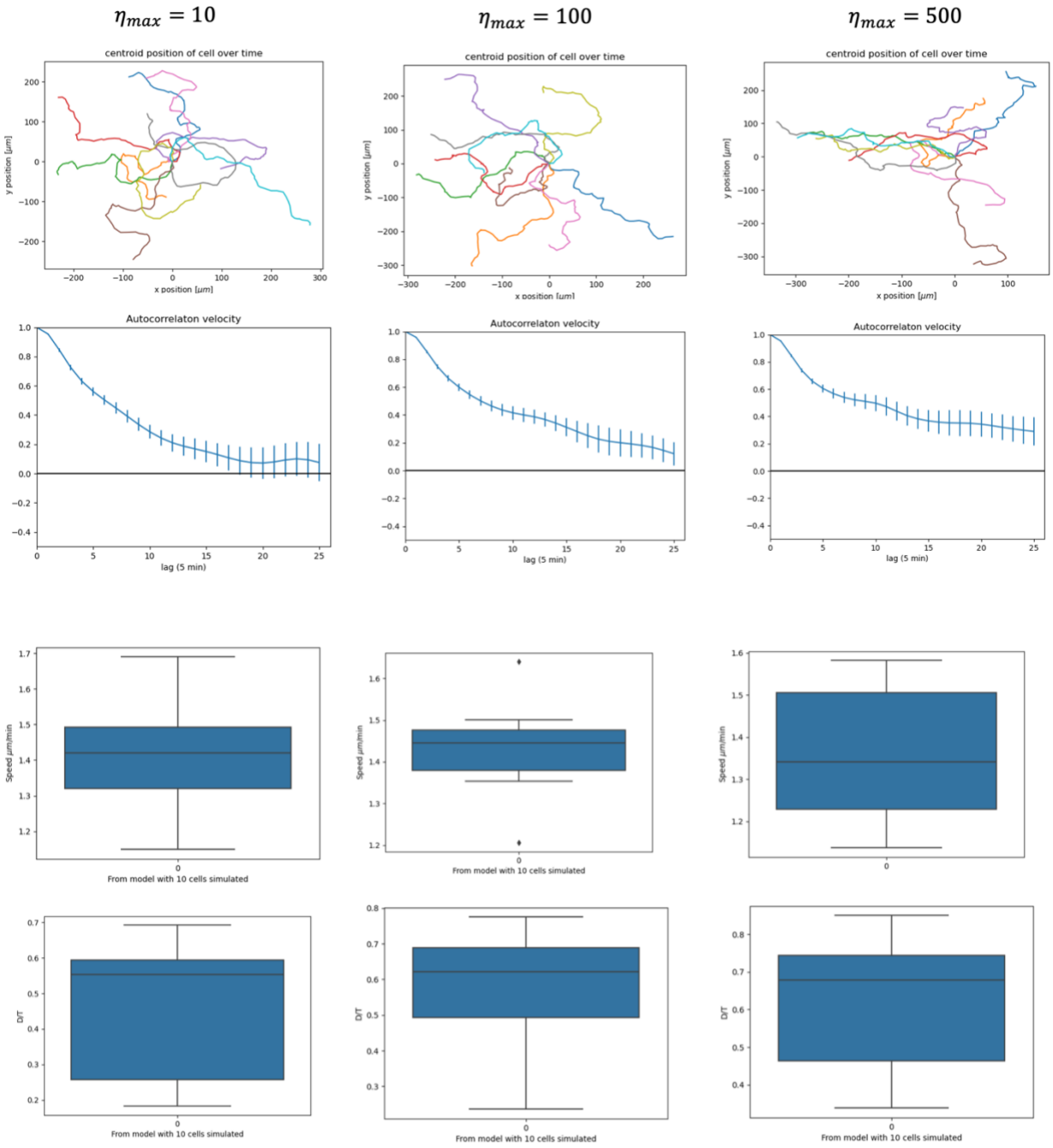


Figure 3.33. Effect of varying η_{max} on a population of cells.

A simulation of 10 cells with the following parameters: $\eta_v^* = 1.67$, $\sigma_\eta = 20, d_{max} = 60$.

Another way to affect the friction coefficient is to change the spatial effect of the internal cell orientation on the friction coefficients at vertices. By increasing σ_η, we increase the effect

cell polarization has on coefficients by increasing the friction at a wider spread vertices opposite to the direction of polarization (Fig. 3.34).

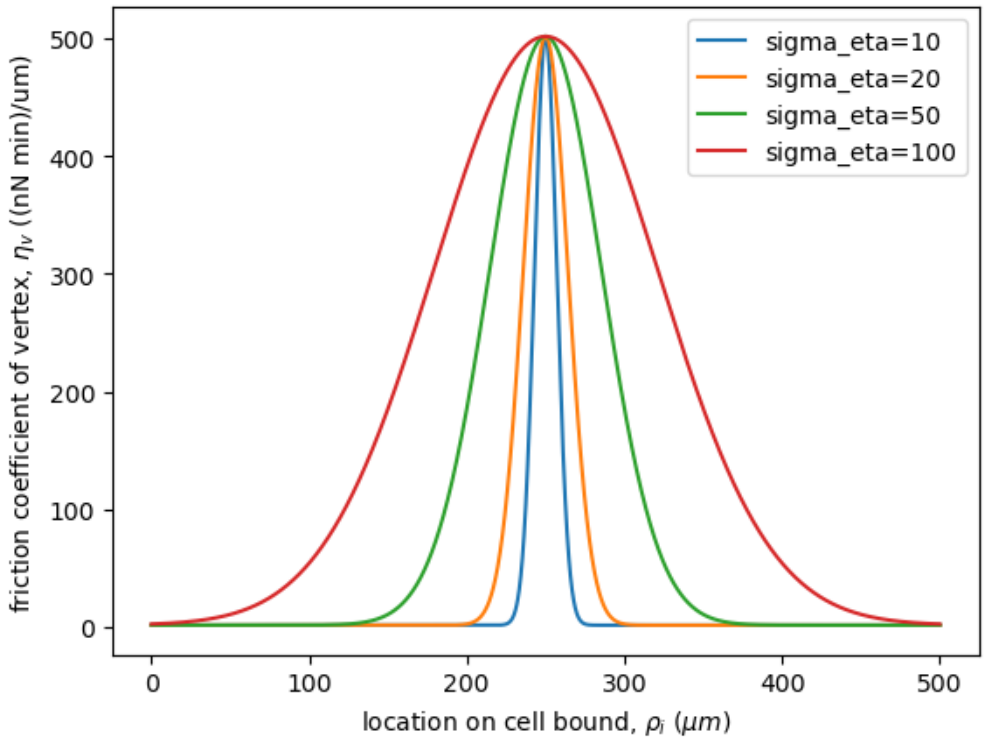


Figure 3.34. Varying σ_η to change the friction coefficient at each vertex, η_{v_i}.
Effect of varying σ_η to change the friction coefficients at each vertex.

By increasing σ_η, cell speed decreases and persistence increases (Fig. 3.35). In this framework, small changes in internal cell orientation has little effect on the friction coefficients of neighboring vertices.

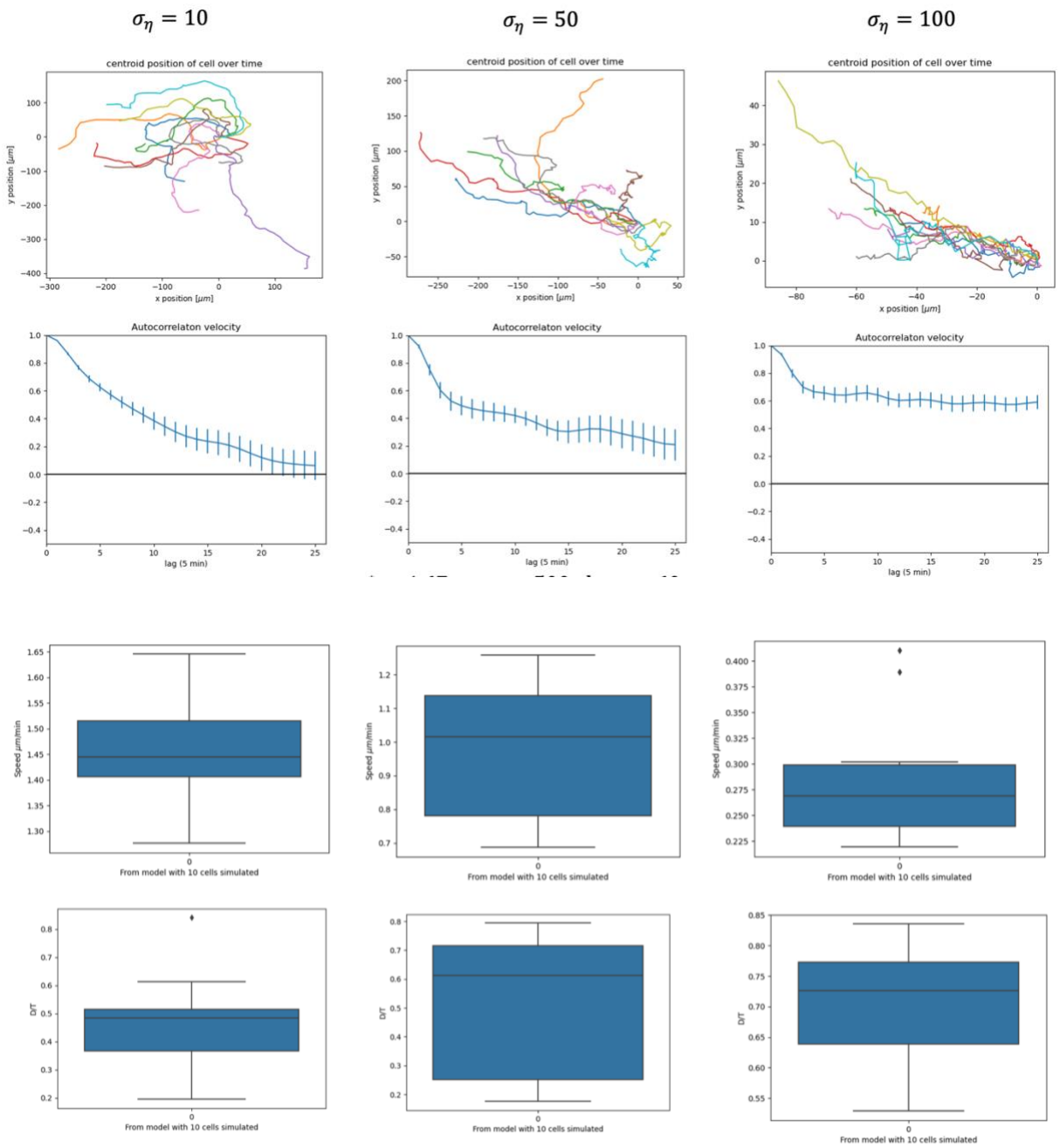


Figure 3.35. Effect of varying σ_η on a population of cells.

A simulation of 10 cells with the following parameters: $\eta_v^* = 1.67$, $\eta_{max} = 500, d_{max} = 60$.

Finally, to vary the friction coefficient, we vary d_{max}, the maximum distance a vertex adhered vertex can be from the center of the cell before becoming unadhered. It seems from the

limited sweep of parameters that varying d_{max} does not have a noticeable change in speed and persistence (Fig. 3.36) or it may be possible that there is an optimal value within the range used that maximizes these values but will require further exploration to confirm.

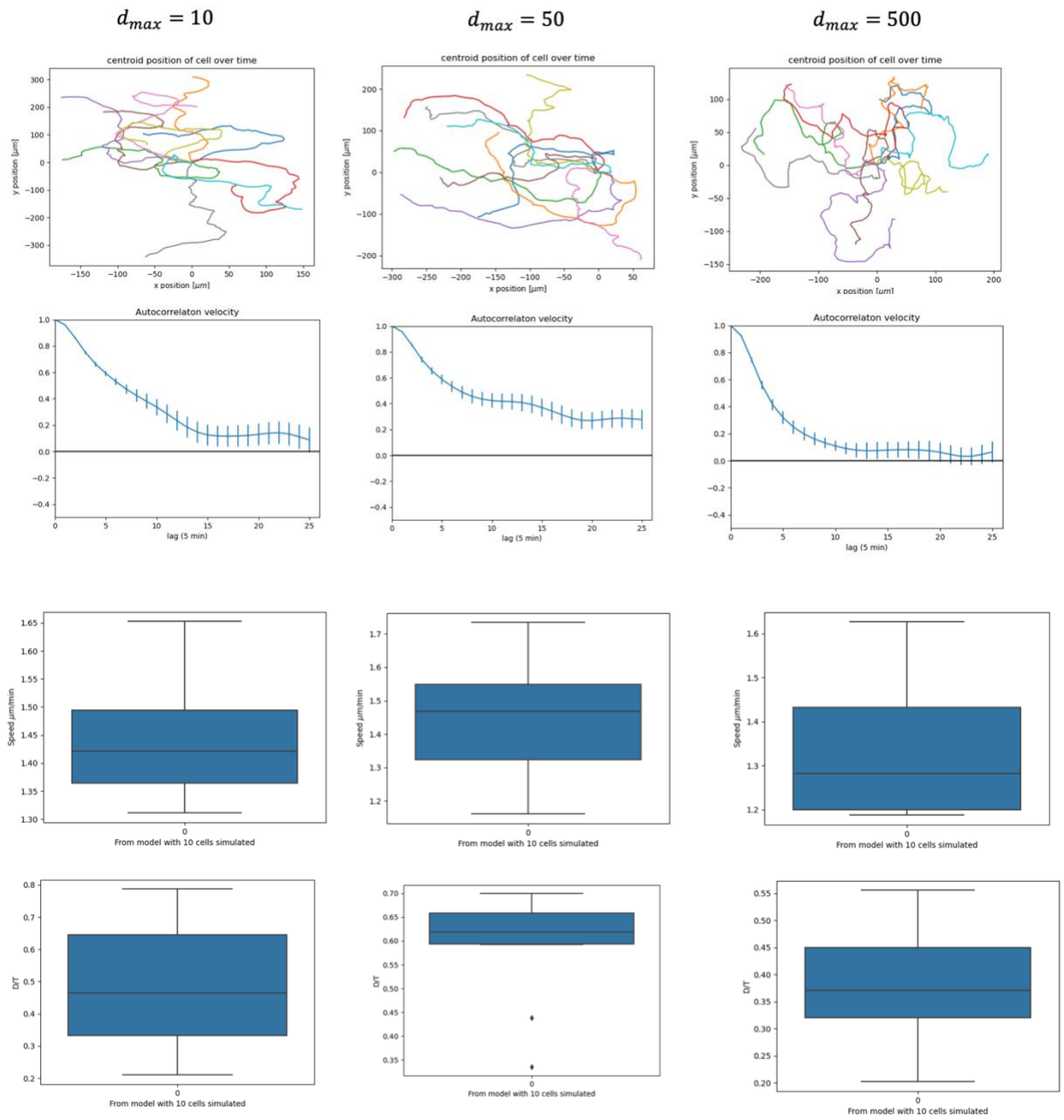


Figure 3.36. Effect of varying d_{max} on a population of cells.

A simulation of 10 cells with the following parameters: $\eta_v^* = 1.67,\ \eta_{max} = 500,\ \sigma_\eta = 20$.

3.5 Future Directions

While we have set up a VBM to spatially model localized activity over the perimeter of the cell to yield overall coordinated cellular movement, the parameters still need to be informed from experimental results. The next steps with this model are to use data to inform the model and build upon the existing model framework.

Using Data to Inform Model and Test Hypotheses

It is important to use real data to inform the model in order to gain meaningful insights into how spatial dynamics coordinate to yield overall cell motility. Fast temporal movies of cell edge dynamics can be used to determine general ranges for k_{on} and k_{off} rates and how protrusion in one region of cell edge effects nearby regions. To adjust the value of k_{on} rates, the values of k_{on}^{*} and T_{max} can simply be changed. To adjust how one area protruding affects another area, we could change T_{max}. To adjust k_{off} rates, we can change k_{off}^{*}, $k_{off_{max}}$ and σ_{off}. To adjust how one area with high off rate (not protruding) affects another region, we can change σ_{off}.

Next, traction force data can be used to determine tension around perimeter of cell and the force dipole major axis as well as the force quadrupole can be used to determine how internal cell orientation changes over time along with giving insight into how distance to center of cell is correlated with tension around the perimeter. For instance: are areas of high outward tension related to concavities as hypothesized by the model (and vice versa high inward with convex regions)? Do these areas of high outward tension correlate to direction of polarization? How

related are shape versus force-based measurement of polarization? Do highest values of force at cell perimeter correspond largest distance to center of cell?

Lastly, by visualizing focal adhesions with time series movies of cells migrating, we can determine if the largest focal adhesions are located opposite to direction of polarization and located at areas furthest from the cell center.

Perturbing Branched Actin

Additionally, we can take advantage of the *Arpc*2 KO cells that have no branched actin to ask if this branched actin is responsible for the "communication" between vertices. We can test this by changing T_{max}. The result of this change would cause protrusion activity to be more localized, thus leading to narrower protrusions, which is what we observe in the data (Figure). This also leads us to ask the question if σ_{off} would also be affected by being smaller, thus making the range of k_{off} narrower. This means that the internal cell orientation would have a more localized effect on turning off protrusive activity in the direction opposite to the orientation. In the data, we see that *Arpc*2 KO cells move more persistently than WT cells (Figure), suggesting that the internal cell orientation updates slower with perhaps a higher friction coefficient.

Perturbing Non-Muscle Myosin II Contractility

We could also use blebbistatin treated cells to inhibit non-muscle myosin II to explore what happens if we affect cell adhesion. In previous experiments, we found that MEFs treated with blebbistatin still were able to migrate despite lacking contractility. These WT cells treated with blebbistatin move faster (Fig. 3.37) and more persistently (Fig. 3.38) than non-treated WT cells. Interestingly, in the model, lower adhesion would result in lower protrusive force rates and

less clearly defined internal cell orientation/polarity. Lowering on rates for the protrusive force increases speed and directional persistence, but a less clearly defined internal cell orientation may decrease directional persistence depending on the friction coefficient for updating this direction. Overall, the current model framework would capture the observed cellular behavior dependent upon the friction coefficient of the internal cell orientation.

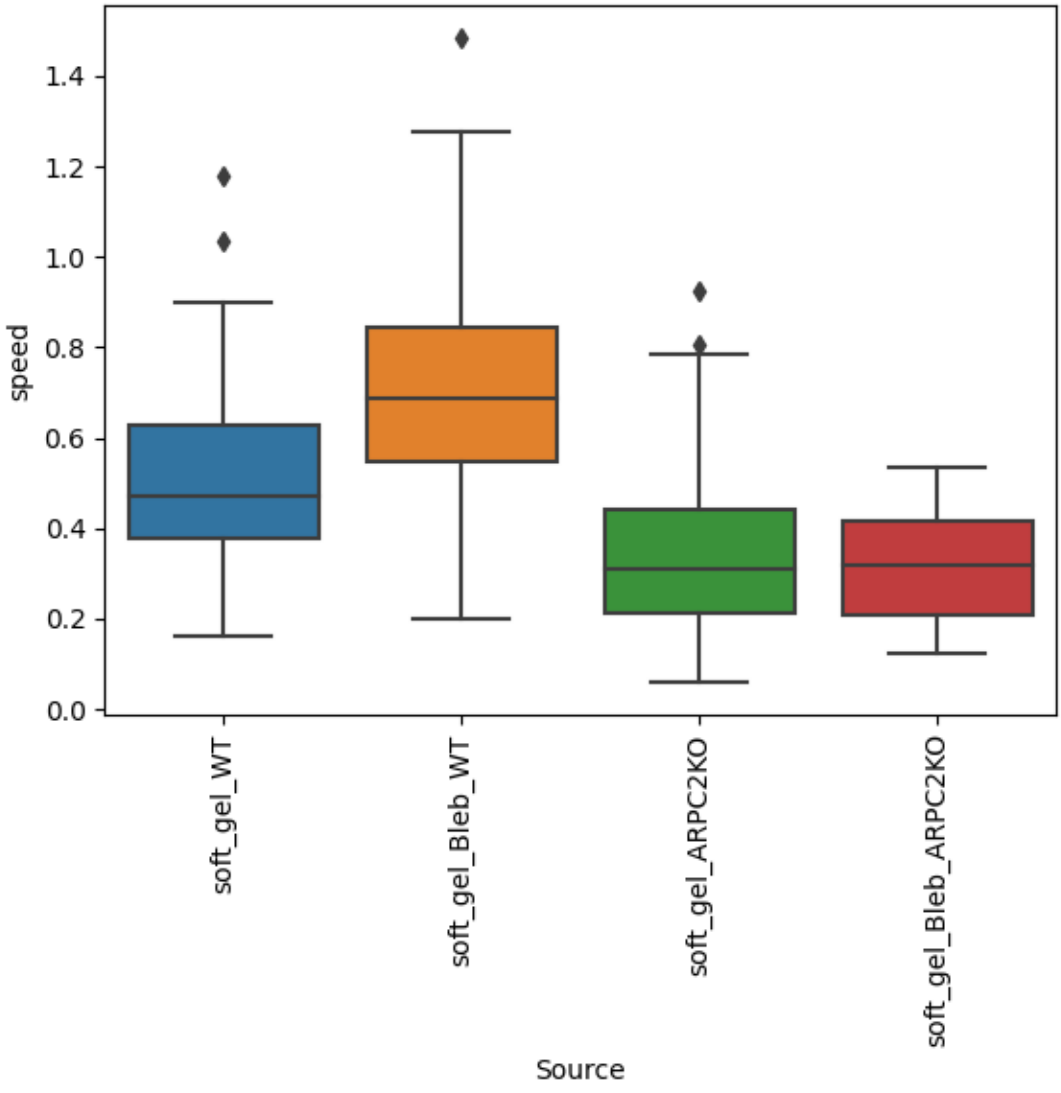


Figure 3.37. Cell speed of WT, *Arpc*2 KO and blebbistatin treated cells.

WT cells treated with blebbistatin display the highest speed amongst the treatments. While the addition of blebbistatin results in faster movement than non-treated WT, it appears to not have an effect with *Arpc*2 KO cells. (140 cells in WT, 78 cells in WT blebbistatin, 149 cells in Arpc2 KO, and 51 tracks in *Arpc*2 KO blebbistatin.

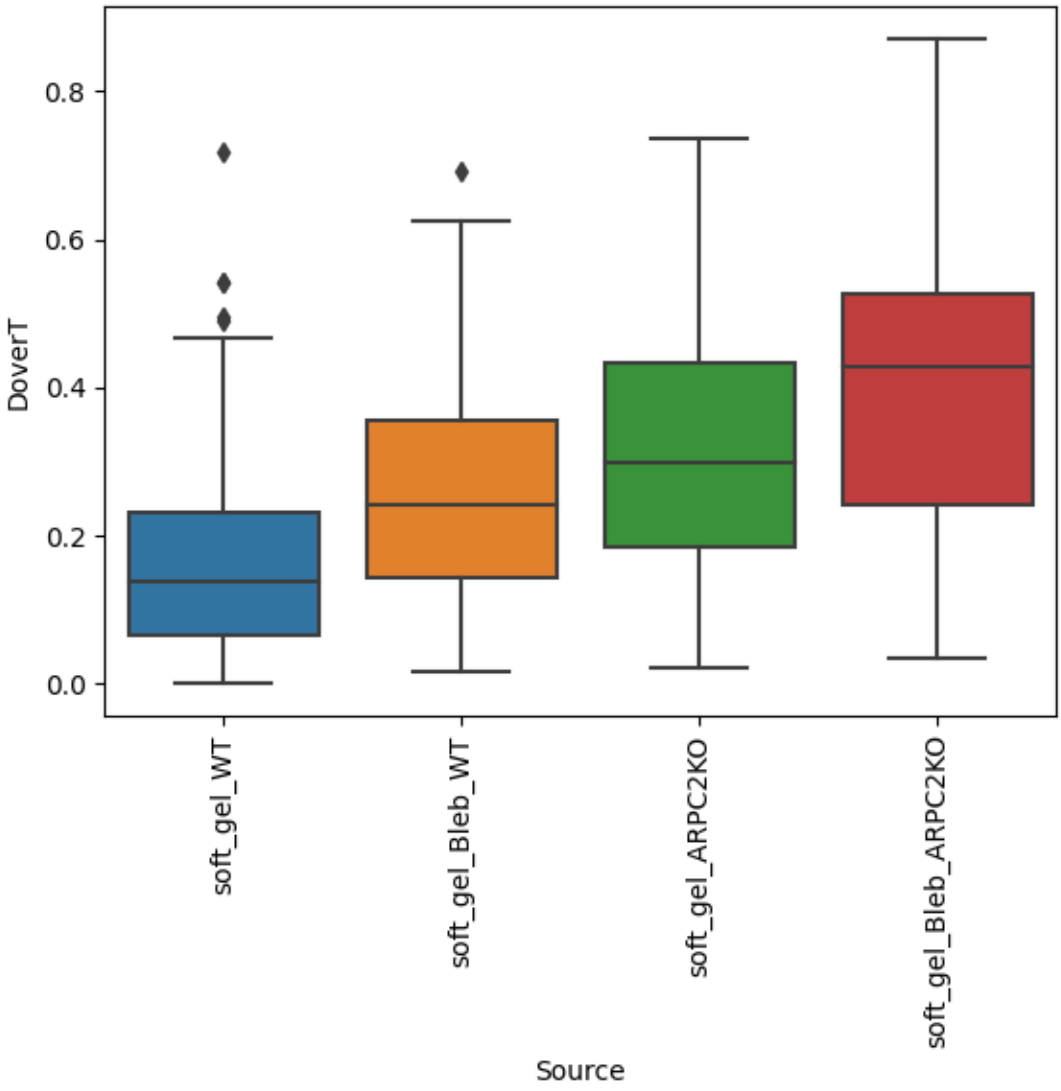


Figure 3.38. Cell persistence of WT, *Arpc*2 KO and blebbistatin treated cells.

WT cells treated with blebbistatin display higher persistence than non-treated WT. However it appears to exacerbate the higher persistence observed in *Arpc*2 KO cells. (140 cells in WT, 78 cells in WT blebbistatin, 149 cells in Arpc2 KO, and 51 tracks in *Arpc*2 KO blebbistatin.

Addition of Contraction

A current limitation of this model is the difficulty of sustaining multiple protrusions over long time periods. It would be worth exploring how can we sustain multiple protrusions in this model framework. A possibility is that contraction may be necessary to yield this behavior. Previously, we had only considered a normal outward protrusive force in the model. However, we would like to add in contraction to make the model more biophysically relevant and perhaps add in the opportunity for multiple protrusions to form. To accomplish this, we now allow the stochastic normal force to act normally inward for contraction or normally outward for protrusion (Fig. 3.39). Contractile forces turn on with rate k_{con}, protrusive forces turn on with rate k_{pr}, and the forces turn off with rates $k_{con\to off}$ and $k_{pr\to off}$, respectively (Fig. 3.40).

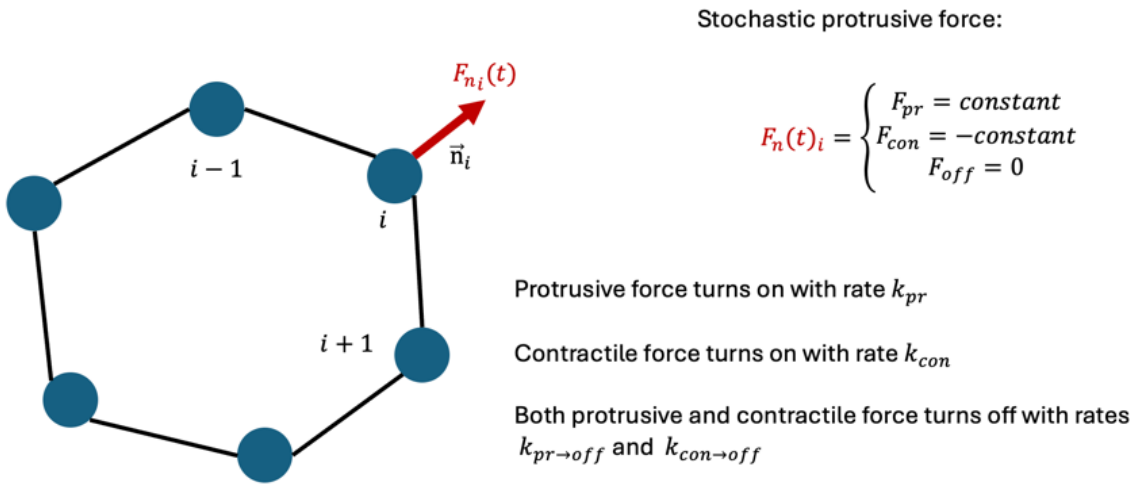


Figure 3.39. Addition of a contractile force.

In addition to a stochastic protrusive force, F_{pr}, that can stochastically turn on and off at each vertex, we can add a contractile force, F_{con}, that can stochastically turn on and off at each vertex. This force acts in the inward normal direction.

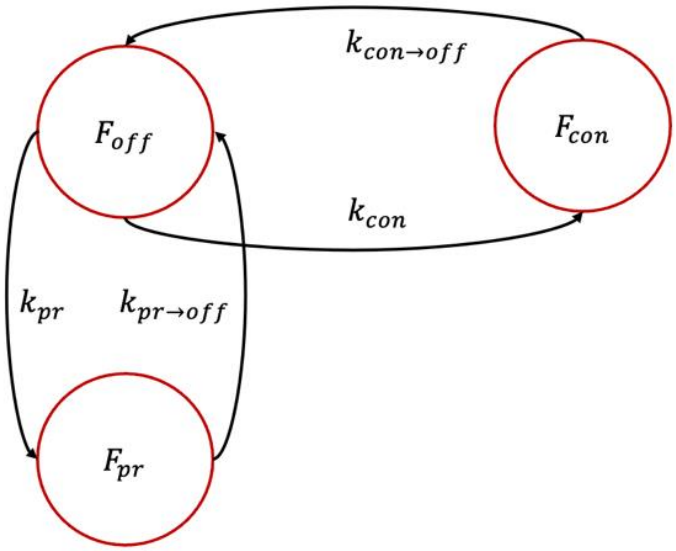


Figure 3.40. Transitions for force states at each vertex.

Each vertex can be in one of three force states: F_{off}, F_{con}, and F_{pr}.

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